# CORRELATIONS OF STUDENTS PREFERENCES ON TECHNICAL-VOCATIONAL-LIVELIHOODTRACK AND ACADEMIC ENGAGEMENT OF COOKERY STUDENTS

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# CHAPTER 1

## Introduction

Academic engagement is commonly defined as the degree to which students are actively involved in their learning through participation, emotional connection, and personal effort in academic tasks (Bond et al., 2020). Academic engagement remains a significant concern among TVL cookery students, as many struggle to stay actively involved in both classroom discussions and hands-on kitchen activities. Low participation, lack of interest in lessons, and incomplete performance tasks are common challenges observed in cookery programs. Several studies were conducted to identify the possible factors that may influence academic engagement, and the majority of the studies suggested that students' preferences for the TVL track are often considered an important factor that may affect how actively and consistently they participate in learning activities. While these studies point to students' TVL track preferences as a potential factor, there remains a limited focus on examining the specific relationship between TVL track preferences and academic engagement among cookery students in senior high schools. This gap provides the rationale for conducting the present study.

Academic engagement issues such as low participation and minimal involvement are not confined to one region. In Ethiopia, Woreta (2024) examined high school students' academic engagement and found persistent problems, including low classroom participation, limited effort in completing tasks, and decreased motivation, even when supportive learning environments were

available. The study also revealed that unclear learning preferences, lack of interest in subjects, and limited guidance from teachers significantly contributed to students' disengagement. Students' personal preferences and contextual factors were suggested as potential influences on engagement without concluding their exact effect. The study recommends conducting further investigations on factors affecting academic engagement, emphasizing the need to explore how specific influences contribute to students' involvement and motivation in learning activities.

In addition, a study in Turkey, Kara and Öztürk (2023) reported that many TVL students show low academic engagement, which refers to the active participation, emotional involvement, and effort students put into their learning activities. This problem often arises because students' preferences on the TVL track are not fully aligned with their chosen program, causing them to feel disconnected from classroom discussions and practical tasks. Traditional teaching methods that ignore students' track preferences can make lessons feel irrelevant, leading learners to participate less, invest minimal effort, and avoid challenging activities, forming a cycle of disengagement that lowers overall academic engagement. The study highlighted students' preferences on the TVL track as a potential influence on academic engagement without concluding the exact relationship. It recommended further research to explore how students' track preferences affect academic engagement in technical-vocational education contexts.

Moreover, a study conducted in Malaysia examined the academic engagement of TVL students and found that many learners struggled to remain

actively involved in both classroom discussions and practical tasks. Students whose program choices did not match their personal interests frequently showed low attendance, minimal participation in practical exercises, and little effort in completing assignments. This mismatch between students' preferences on the TVL track and the curriculum appeared to demotivate learners, lowering their overall academic engagement (Hassan & Jamaludin, 2022).

Nationally, in Iloilo City, Olaguer and Pelayo (2023) reported that senior high school TVL students often face difficulties in performance assessments, which can affect their academic engagement during skill evaluations. Many students described experiences of frustration and worry during practical performance tasks, especially when they felt unprepared or when assessment criteria did not match their personal learning preferences. These assessment challenges resulted in stress, reduced confidence, and occasional withdrawal from actively trying new techniques. The study highlighted students' preferences on the TVL track as a contextual influence on engagement without concluding the exact nature of the relationship. It recommended further research to examine how students' track preferences interact with performance assessment experiences to shape academic engagement.

In addition, in Bohol Province, Alejandro and Bucad Jr. (2024) found that although many senior high school TVL students show overall readiness for vocational programs, gaps in certain skill domains such as communication and technical adaptability influenced their engagement in complex learning tasks. Students reported that when their preferences on the TVL track skewed toward

more creative or hands-on models, they became less engaged in activities requiring strong theoretical understanding or precision. These mismatches between preference and task demands appeared to reduce students' willingness to invest deeper effort in some learning components.

Moreover, in Cebu City, Philippines, Blanco and Tingzon (2024) observed that senior high school students' reasons for choosing the TVL track, such as parental influence, career expectations, and peer trends, played a role in how they engaged academically once enrolled. Students who selected the TVL track mainly due to external influences rather than their own preferences reported feeling disconnected from practical cookery lessons and theoretical discussions, and were more likely to struggle with maintaining consistent effort and involvement. The study introduced students' preferences on the TVL track as a potential context affecting academic engagement without concluding the relationship.

Locally, in Tagum City, Davao del Norte, Philippines, Basaya and Bautista (2024) reported that senior high school TVL students faced challenges in completing technical writing and documentation tasks, which affected their academic engagement in school activities. Many students expressed difficulty in organizing ideas, following technical formats, and meeting deadlines, particularly when the tasks did not align with their personal preferences on the TVL track. These challenges led to increased stress, lowered confidence, and occasional avoidance of more complex academic tasks. The study highlighted students' preferences on the TVL track as a contextual factor that may influence academic engagement without establishing a direct causal effect. It recommended further

research to investigate how alignment of students' track preferences with learning tasks could improve engagement and performance in senior high school TVL programs.

**Rationale of the Study**

The researchers will carry out this study to find out if there is an important connection between student's preferences and the academic engagement of senior high school students. The purpose of the study is to look into the ongoing problems of low and inconsistent academic results and to explore how students' ways of studying might affect these issues. Based on what was learned from related readings and the challenges seen among students, the researchers believe it is important to do this study to better understand how student preference influence students' overall engagement, especially among senior high school students in Tagum City. It is also necessary to fill this gap in knowledge to offer proof that can help create better ways to improve learning outcomes**.**

*School Administrators.* This study will help administrators better understand how students' preferences on the TVL track influence the academic engagement of TVL cookery students. This data will allow them to plan programs, organize activities, and allocate resources that support student-centered learning and foster higher participation and involvement in both practical and theoretical lessons.

*Department of Education*. The findings of this study will provide valuable data on how students' preferences on the TVL track affect the academic engagement of TVL cookery students. This information may serve as a basis for developing programs and policies that align TVL track offerings and learning

activities with students' interests, helping to improve participation, motivation, and overall engagement in senior high school cookery classes.

*Teachers.* Teachers will benefit from this study by gaining a clearer understanding of how student preferences affect academic engagement. This knowledge can help them design teaching strategies that not only present lessons effectively but also support students in becoming more involved and interested in learning activities.

*Parents.* Parents may also benefit, as the findings can help them understand the importance of supporting their children in choosing learning methods that suit their preferences. They can learn practical ways to guide and encourage their children's active participation in school activities, which may lead to better academic engagement.

*TVL Students.* Students stand to gain the most from this study. By understanding how their learning preferences influence their level of engagement, they can make better choices in how they approach school activities. This awareness may encourage them to move away from passive or less effective habits and become more actively involved in learning through methods that suit their interests and strengths.

*Future Researchers*. Future researchers will benefit from this study as it offers a useful foundation for exploring the link between student learning preferences and academic engagement. It can serve as a reference for developing related studies, refining research methods, or expanding the topic to other educational settings. This study also highlights areas that need further

investigation, encouraging future scholars to deepen the understanding of how personalized learning can enhance student outcomes.

**Statement of the Problem**

1. What is the level of student preferences on TVL track of cookery students in terms of:

1.1 Personal Factors

1.2 Parents Encouragement

1.3 Teachers/Classmates/Friends Encouragement

1.4 Church encouragement

1.5 Opportunity Factor

1.6 Preferred Career Choices?

2. What is the level of academic engagement cookery students in terms of:

2.1 Interests

2.2 Study Habits

2.3 Academic ability

2.4 Teacher related aspect

2.5 On-Instructional Materials

3. Is there a significant relationship between student preferences on TVL track and the academic engagement of the cookery students?

**Null Hypothesis**

The study will be tested at 0.05 significance

H0: There is no significant relationship between Student Preferences on TVL Track and student Academic Engagement among Cookery students.

**Review of Related Literature**

The researchers offer a thorough review of relevant literature in the field, helping readers build a clearer understanding of the study's main components. They begin by introducing the independent variable, carefully outlining its indicators to show how it functions within the research framework. Following this, they shift focus to the dependent variable, discussing its role and significance while connecting it to previous studies that support and contextualize the current investigation.

**Student Preferences on Technical-Vocational Livelihood Track**

Student Preferences on TVL Track refers to the degree to which students show interest or inclination toward specific strands or subjects within the Technical-Vocational-Livelihood (TVL) track, influenced by personal interest, parental encouragement, teachers' and classmates' support, community or church guidance, perceived opportunities, and preferred career choices (Salvador et al., 2021). It is operationally defined in this study as the measurable inclinations of TVL students toward their chosen strand, assessed through indicators such as personal interest in the track, parental encouragement, teachers' and classmates' influence, community or church support, awareness of opportunities, and alignment with preferred career choices.

An international quantitative study conducted in Indonesia investigated how technical-vocational education (VE) students form career preferences within

vocational education programs and found that self-efficacy significantly shapes students' career choices toward vocational pathways. Specifically, the research showed that students with higher perceived digital competence and psychological well-being were more likely to prefer and commit to vocational education choices, underscoring the role of internal belief systems in track preference formation. This study suggests that strengthening students' confidence and competencies early may influence greater interest and alignment with vocational education offerings such as TVL in senior high schools (Kholifah et al., 2025).

Conversely, a cross-national investigation into vocational and academic pathway decisions among secondary school students in Chile revealed that students' choices between vocational and general educational tracks were driven by perceived relevance to future employment and personal values, with those emphasizing hands-on vocational skills expressing stronger preferences for vocational routes over purely academic tracks. The study illustrates that when vocational education is seen as aligned with practical goals, students' preference for such tracks increases (Blazquez, 2024).

Meanwhile, research in Europe examining differentiated choices within vocational education highlights that students' values associated with specific vocational fields contribute substantially to their preferences for vocational education paths, with distinct value systems explaining why students opt for certain vocational orientations over others. The findings indicate that preferences in vocational educational choices are not uniform but shaped by localized value interpretations and career goal alignment (Delay, 2022).

In the Philippines, a study in Davao Del Sur found that Filipino students' intention to enroll in the Technical-Vocational-Livelihood (TVL) track is significantly influenced by their own preferences shaped by social context, with students more likely to choose TVL when family and classmates show positive attitudes toward such programs. The study concluded that career counselors should enhance awareness and alignment of student preferences to improve TVL enrollment decisions (Penduko, 2021).

Additionally, a quantitative investigation focusing on senior high school track choices in a Quezon City public school determined that approximately 25 % of respondents expressed a clear preference for the TVL track, valuing its practical, skills-based orientation and potential for direct employment or entrepreneurship after graduation. The results reflect a discernible segment of Filipino students prioritizing vocational track preferences in response to personal goals and labour market expectations (Sison, 2024).

Furthermore, a study in Batangas found that graduates reported specialization preferences within TVL fields that were aligned with career objectives; for example, many chose cookery as a preferred TVL specialization. This descriptive insight suggests that TVL students make deliberate decisions about their vocational focus based on both interest and perceived relevance to their future pathways (Valerio & Accad, 2025).

A local study conducted among senior high school TVL students in Panabo assessed readiness but also implicitly indicated that students' comfort and preparedness in skills-oriented domains reflect underlying preferences for TVL

education. Although the primary focus was readiness, the research notes that students' learning experiences and enthusiasm for hands-on vocational training reveal an inherent preference for TVL tracks that offer practical and technical engagement. This highlights how local TVL student populations value experiential learning aligned with vocational track structures (Alejandro & Bucad Jr., 2024).

*Personal factor.* Personal factor refers to the individual traits, characteristics, and preferences that influence a student's decisions and inclinations toward educational tracks or career paths. It encompasses students' interests, motivations, self-efficacy, and personal goals, which collectively shape their evaluation and selection of opportunities such as the Technical-Vocational-Livelihood (TVL) track (Sanchez & Ramos, 2024). In this study, personal factor is operationally defined as the student's own preferences and motivations that guide their choice of the TVL track, measured through self-reported questionnaires assessing interest, alignment with personal goals, and perceived suitability of the track.

Internationally, a study in Pakistan indicated that about one-third of students in surveyed areas preferred vocational tracks such as TVL, particularly when their own personal interests and inclinations toward practical learning were strongly aligned with the hands-on nature of vocational curricula. Consequently, the study suggests that intrinsic preference plays a dominant role in deciding on vocational pathways, as students tend to gravitate toward tracks that resonate with personal inclinations (Ali, 2021).

Similarly, research in Switzerland examining the determinants of educational track selection showed that students' vocational interest dimensions as conceptualized in Holland's model of career choice significantly predicted enrollment in vocational education pathways, with those whose personal interest profiles aligned with vocational tasks more likely to choose vocational education over general or hybrid tracks; this highlights the importance of personal vocational interests in educational preferences (Burkhauser & Weiss, 2021).

Moreover, a study in Germany on choices between academic and vocational schooling found that students' expressed vocational interests were significant predictors of opting for vocational education tracks, with practical interest patterns significantly associated with vocational track selection, suggesting that participating countries' data reflect how personal preference for vocational learning environments shapes educational choices (Ackerman & Heggestad, 2020).

In the Philippines, a study within the Division of Toledo City reported that student's personal interest was one of the most influential reasons for preferring vocational pathways such as TVL, with students indicating that their own preference for task-related activities guided their decisions in choosing their preferred strand (Doctolero, 2020).

Likewise, research conducted among Bulacan State University senior high students showed that students' own interests were the most important factor in deciding to enter the TVL track, as learners themselves cited personal liking for

technical and livelihood learning as a primary reason for their choice because they prefer doing activities than academics and homework (Dalisay, 2021).

Furthermore, findings from a descriptive study in Tacloban of incoming senior high school students revealed that a substantial number preferred the TVL track because they believed it best suited their personal choices, interests, and childhood aspirations, demonstrating that students' own preference is a critical driver in TVL track selection and preference (Rodriguez, 2021).

Lastly, a local investigation in Compostella Valley indicated that while multiple predictors influence TVL track choice, the focus on personal preference was emphasized by respondents as a key internal reason for opting for the TVL track, highlighting that, in this locality, students' own personal choice and interest significantly shaped their decision to prefer the vocational pathway (Capiz & Malayas, 2020).

*Parents' encouragement.* Parents' Encouragement refers to the emotional, motivational, and supportive actions taken by parents that encourage their children to pursue certain educational pathways or career decisions, including offering positive reinforcement, verbal affirmation, and approval of a student's interests and choices (Shin et al., 2024). In this study, parents' encouragement is operationally defined as the extent to which senior high school students perceive their parents actively motivating, supporting, or affirming their decision to choose the Technical-Vocational-Livelihood (TVL) track, measured through self-reported questionnaire items assessing perceived parental support and approval in the process of TVL track selection.

An international investigation in Oyo State, Nigeria studies the relationship between parental support and students' choice of vocational and technical career pathways found a significant positive association between parents' encouragement and students' likelihood to choose technical and vocational studies, as higher levels of parental support were linked to greater selection of these programs among transitional students indicating that when parents actively encourage technical pathways, students are more inclined to pursue them (Adegbite et al., 2024).

Furthermore, research conducted in Punjab, Pakistan on secondary students' attitudes toward technical and vocational education reported that a majority of respondents agreed that parental influence significantly affects students' attitudes and willingness to join vocational programs, suggesting that when parents exhibit encouraging behaviour or express positive views about vocational training, students are more receptive to choosing vocational fields (Lavendets et al., 2022).

In addition, literature on vocational development in North America contexts emphasized that parents play a stronger role than peers or schools in shaping students' vocational interests and encouraging them toward vocational choices, demonstrating that earlier parental involvement and support are crucial in students' decisions to consider vocational pathways, including technical studies, in upper secondary settings (Boonk et al., 2021).

At the national level in the Philippines, research synthesizing factors influencing senior high school track preference reported that parental guidance

and support were important influences on students' decisions, with those having supportive parents being more likely to select their preferred track, including vocational ones such as TVL, illustrating that parents' encouragement helps affirm and reinforce students' preferences (Sasan, 2023).

Likewise, a descriptive document reviewing factors related to TVL career choices among Philippine senior high students highlighted that students who perceived positive parental influence regarding career options were more confident in their choice of vocational pathways, indicating that parents' encouragement can bolster students' decisions to pursue TVL, especially in regions where family guidance is valued (Ambajao, 2020).

Moreover, a study in Loon, Bohol examined senior high school track preferences in the Philippines found that peer and parental support were linked to students selecting their preferred educational pathways, with parents' encouragement specifically associated with higher willingness among learners to choose vocational tracks that matched their interests and aspirations (Lumboy, 2019)

Finally, a study from Samal Island explored parental perceptions and students' career decisions in  as an example of localized context within broader vocational research  suggested that positive encouragement from parents significantly correlates with students' willingness to pursue vocational careers, a finding that implies in local settings where parents actively encourage vocational pathways like TVL, students' preferences for such tracks are strengthened (Kabuyao, 2025).

*Teacher/Classmates/FriendsEncouragement.*

Teachers/Classmates/Friends Encouragement refers to the verbal support, motivational feedback, positive reinforcement, and guidance provided by teachers, classmates, and friends that influence students' academic motivation, engagement, and educational decision-making. In vocational education settings, perceived teacher support has been associated with higher levels of learning engagement and improved academic outcomes, while peer encouragement has been shown to strengthen academic adjustment and sustained participation in school activities. Such social reinforcement enhances students' confidence, persistence, and clarity in choosing educational pathways, including vocational tracks like the TVL strand, and provides emotional security that helps them manage academic challenges and remain focused on their goals (Jansen et al., 2025). In this study, Teachers/Classmates/Friends Encouragement is operationally defined as the extent to which students perceive and report receiving positive verbal motivation, academic guidance, approval, and social support from their teachers, classmates, and friends regarding their preference for the TVL track, as measured through specific items in the survey questionnaire.

In an international setting, a study investigated the relationship between students' liking of teachers and learning engagement among vocational college learners in China and reported that teachers' support and student-teacher rapport positively influenced engagement. Their findings suggest that encouragement and positive interactions from teachers increase students' emotional investment and persistence in learning, which can directly impact their academic choices and

engagement in vocational programs. The study further emphasized that consistent teacher encouragement not only improves participation in class but also fosters students' intrinsic motivation to excel academically (Lu et al., 2022).

Another international study analysed teacher and parent support on academic achievement of secondary vocational students in Shanghai, China, finding that autonomy, emotional, and academic support from teachers significantly correlated with students' academic performance and engagement. This indicates that teacher encouragement whether emotional or instructional can shape students' orientation toward continued study and educational choices. Moreover, the study highlighted that supportive teachers help students develop self-regulation skills and confidence, which are critical for success in vocational learning environments (Peng et al., 2022).

Furthermore, a study investigated the influence of perceived peer support on academic adjustment among higher vocational college students across China and found that perceived peer encouragement and support significantly predicted students' academic adjustment and engagement, partly through mediators like academic hope and professional identity. This demonstrates the critical role of classmates and friends in enhancing vocational learners' motivation and academic engagement. The findings also suggest that positive peer networks provide both informational and emotional support, reinforcing students' sense of belonging and commitment to their chosen tracks (Zhu et al., 2025).

In a national setting, a study on Philippine SHS track determinants noted that peer influence and encouragement from classmates were significant factors

affecting senior high school students' decisions about educational track preference. Peer encouragement was cited alongside parental and academic support as noteworthy influences on how students decide between different pathways. This highlights that even in localized contexts, classmates' reinforcement plays a key role in guiding students toward vocational or academic tracks (Kilag et al., 2023).

Additionally, a study identified that classmates and peers are important sources of career guidance, with students exchanging career information and encouragement in ways that affect their decisions on which tracks to pursue, including vocational options. Peer interactions often serve as informal mentoring, allowing students to share experiences and advice that inform practical decisions about education. Consequently, these peer networks become influential in shaping students' understanding of career pathways and future aspirations (Salingsing, et al., 2021).

Furthermore, a study found that peer support significantly influences career decision self-efficacy, meaning that encouragement from friends helps students build confidence in making their own career choices rather than simply following trends, which is vital when deciding on TVL tracks. This implies that students who feel supported by peers are more likely to engage in reflective and independent decision-making regarding their academic and vocational pursuits. Peer encouragement therefore not only supports motivation but also strengthens students' autonomy in career planning (Manaligod, et al., 2022).

A local study on senior high school graduates in Davao City found that peer interactions and encouragement influenced graduates' career decisions, including preferences for technical or vocational pathways. Students who received positive reinforcement from classmates were more likely to consider vocational programs as viable post-secondary options, indicating that encouragement from peers significantly affects preferences in this regional context. The study further noted that such encouragement often interacts with family and teacher guidance, creating a supportive ecosystem that enhances the likelihood of pursuing TVL tracks (De Leon, 2025).

*Community/Church encouragement.* Community/Church Encouragement refers to the support, guidance, and motivation students receive from community members or church groups that influence their academic motivation and educational decisions. Studies indicate that students who receive strong community or faith-based encouragement demonstrate higher academic performance and stronger learning motivation, which are linked to persistence and clearer educational pathway choices, including vocational tracks (Guanlo et al., 2024). In this study, Community/Church Encouragement is operationally defined as the extent to which students perceive and report receiving motivational support and positive reinforcement from their community or church regarding their preference for the TVL track, as measured through survey items in the questionnaire.

In Bangkok, Thailand, a study conducted at a faith-based university investigated how encouragement to engage in religious and moral activities relates

to students' academic performance and engagement. The findings showed that students who received guidance or motivation to participate in structured faith-based events achieved higher academically. Thus, support from community or church activities strengthens students' discipline, moral values, and commitment to education, which are directly associated with their preferences and engagement in vocational tracks (Kitjaroonchai & Hungyo, 2020).

In Addis Ababa, Ethiopia, research examined the relationship between encouragement for community engagement and academic achievement in higher education institutions. The results revealed that students who were supported or motivated to actively participate in community programs demonstrated higher academic performance, stronger learning motivation, and greater persistence in their studies. Accordingly, these findings imply that encouragement cultivates attitudes and competencies that contribute to academic success and influence students' preferences for specific pathways such as Technical-Vocational-Livelihood programs (Belete, 2024).

Across higher education institutions in the United States, a study on service-learning programs integrated into academic curricula assessed the effect of encouragement for community participation on student engagement. The findings indicated that students who were motivated or supported to engage in community projects perceived their coursework as more meaningful and aligned with real-world applications. Consequently, this affirms that encouragement not only promotes civic and personal development but also reinforces academic

commitment and supports career-oriented educational decisions (Holmes et al., 2022).

Within the Philippine context, a study conducted at a Catholic higher education institution in Bacolod City, Negros Occidental, Philippines examined awareness, encouragement, and motivation for community outreach. The findings disclosed that although awareness of community programs was high, actual student participation differed according to the degree of encouragement and motivational support provided. In effect, this points to the significant function of community and church encouragement in strengthening students' values formation, active involvement, and dedication to educational initiatives (Calimpos et al., 2023).

Correspondingly, in Ozamiz City, Philippines, a study analyzed the relationship between encouragement to participate in church activities and values development. The results demonstrated that students who were urged to engage in church programs manifested stronger development of values such as composure, peace, and tolerance. In light of this, church encouragement contributes to personal and moral advancement, which subsequently reinforces broader academic engagement and sustained learning habits (Jomuad et al., 2021).

Along similar lines, a study involving senior high school students at Jomalig National High School in Quezon Province, Philippines emphasized the role of encouragement from parents, schools, and communities. The findings indicated that students who were consistently motivated to take part in social and community

activities showed enhanced academic performance and engagement. Taken together, these outcomes affirm that encouragement reinforces students' commitment, motivation, and overall educational attainment (Florez & Perez, 2022).

Locally, a mixed-methods study conducted at Mangguangan Integrated School in New Corella, Davao del Norte, Philippines explored students' experiences in school organizations and activities. The study found that encouragement and support to participate in these organizations significantly strengthened students' leadership, responsibility, and commitment. These results suggest that community and church encouragement positively influences students' academic and extracurricular engagement, helping them develop habits that support both educational and communal goals (Azua & Sario, 2024).

*Opportunity factor.* Opportunity Factor refers to students' perception of future employment prospects, career advancement, and economic benefits associated with a specific educational track, which influences their track preference and decision-making. In educational research, it is commonly examined through indicators such as expected job stability, income potential, employability, and alignment with labor market demands. Studies indicate that when students perceive stronger economic and career opportunities linked to a pathway, they are more likely to prefer and commit to that track, particularly in technical and vocational programs (Nguyen et al., 2023). In this study, Opportunity Factor is operationally defined as the extent to which students perceive and report

that the TVL track provides favorable job prospects and income potential, as measured through survey questionnaire items.

In an international context, a study conducted in Switzerland demonstrated that anticipated career benefits including employment prospects and perceived future opportunities substantially shape secondary students' intentions toward technical and engineering careers. Consequently, these opportunity perceptions, influenced by local industry engagement programs, highlight that job prospects and practical career alignment are essential determinants of students' educational choices and, by extension, their engagement in those tracks (Smit et al., 2020).

In a comparable vein, research in Spain explored vocational values among students with interests in fields such as sport science. Although the study did not explicitly use the term "opportunity factor," its emphasis on vocational values—particularly regarding professional development and work-life considerations—implies that perceived opportunities, including employment potential and job satisfaction, inform students' educational preferences. Thus, opportunity-related considerations play a meaningful role in shaping both track selection and academic wellbeing internationally (Franco et al., 2021).

Correspondingly, a study in China found that factors such as available work opportunities, family income, and individual aptitude directly influenced senior high school students' track choices. Importantly, future employment prospects emerged as a decisive element, with students favoring tracks that promised clear pathways to viable careers. Accordingly, these findings reinforce the international pattern that

perceived labor market alignment and career opportunities strongly guide students' academic engagement and vocational track preferences (Royo & Lamela, 2021).

Across the Philippines, a study investigating factors influencing Senior High School students' career track choices across multiple regions, with a large sample of Grade 11 learners, revealed that students' anticipated future careers and expectations—especially regarding potential job opportunities linked to specific tracks—significantly affected their decisions. Consequently, these findings suggest that considerations of future employment and career prospects guide students' educational preferences, with opportunity-related factors influencing selections for tracks like Technical-Vocational-Livelihood and shaping engagement in pathways perceived to offer desirable outcomes (Nazareno et al., 2021).

In a related case, research at Saint John Berchmans High School in Isabela, Philippines, found that job opportunities served as a moderately influential determinant in students' strand selection. Participants reported that anticipated work environment and job stability impacted their choices. Therefore, even when personal interests and family influences are considered, opportunity factor—measured as expected career prospects—remains a significant influence on students' educational decisions at the national level (Sabanal, 2025).

Likewise, a descriptive-correlational study among Grade 10 students in Pamplona Cluster III, Philippines, examined factors affecting track and strand preferences, including perceived job opportunities. Results indicated that students factored in long-term employability and potential future work when making selections, highlighting that opportunity considerations are integral to decision-

making. Hence, expectations of employment and career alignment strongly shape Filipino students' preferences for technical and vocational tracks such as TVL, consistent with broader K-12 education research (Calibugar, 2025).

Moreover, at a local level in Cotabato Province, a study exploring enrollment in Career and Technical Education programs identified socioeconomic and opportunity-related factors as key influences. Respondents emphasized that these programs provide advantages in terms of employment relevance and skill acquisition for future work. Accordingly, in the Mindanao context, perceived opportunities play a decisive role in guiding students toward vocational tracks, including TVL fields such as cookery (Cataguran, 2025).

*Preferred career choices.* Preferred Career Choices refer to students' expressed interest in specific occupations or professional pathways based on their interests, perceived abilities, values, and anticipated opportunities. In TVL tracks such as Cookery, these preferences influence motivation, persistence, and engagement in skill-based training, as well as academic behaviors and long-term vocational decisions. Studies have shown that career interests significantly predict students' educational engagement and career planning behaviors (Lent et al., 2020). In this study, Preferred Career Choices are operationally defined as the extent to which students report favoring careers related to the TVL track, measured through survey items assessing their interests, perceived competence, and expected opportunities.

Internationally, a study conducted in Australia explored secondary students' vocational preferences and academic engagement in career-oriented programs.

Notably, the findings revealed that students with clearly defined career preferences demonstrated higher persistence and more active participation in practical coursework. Consequently, the study emphasized that alignment between preferred career paths and enrolled programs strengthens motivation and skill mastery in technical education (Howard et al., 2021).

In a similar investigation, research in Indonesia examined vocational high school students' career aspirations and learning engagement. Importantly, students who expressed strong preferences for hospitality and culinary careers achieved higher academic performance in practical subjects. Therefore, the study concluded that career preference serves as a significant predictor of engagement in technical-vocational coursework (Suryadi et al., 2020).

Likewise, a study in Turkey investigated the connection between career decision-making and academic motivation among technical education students. Significantly, students with well-defined occupational preferences demonstrated stronger engagement and commitment to skill acquisition. Hence, the findings underscore the importance of supporting students' career clarity to enhance vocational learning outcomes (Korkmaz & Altun, 2020).

Nationally, a study conducted in Metro Manila examined Senior High School students' TVL track preferences and academic engagement. Notably, students who intentionally chose the Cookery specialization based on career aspirations demonstrated higher practical performance and classroom participation, highlighting that career-driven choices enhance engagement in technical-vocational subjects (Baluyos et al., 2020).

Similarly, research in Central Luzon investigated factors influencing Senior High School students' selection of TVL strands. Importantly, career preference emerged as a significant predictor of academic motivation and sustained engagement in skills training, with students who selected their strand based on long-term employment goals showing higher commitment to coursework (Tan, 2023).

Likewise, a study in Cebu City assessed the relationship between career aspirations and academic performance among TVL students. The findings revealed that learners with strong culinary career goals performed better in competency-based assessments, indicating that preferred career choices positively influence engagement and achievement in vocational education (Oducado et al., 2020).

At a local level in Davao del Norte, research examined TVL strand preferences and academic engagement in public secondary schools. Significantly, students who chose Cookery due to clear employment goals in the hospitality industry demonstrated higher participation in laboratory activities and practical assessments, underscoring that aligning preferred career choices with enrolled specializations enhances both engagement and skill competency development (Avestruz, 2021).

**Academic Engagement**

Academic engagement is defined as a multidimensional construct that encompasses students' behavioral participation, emotional involvement, and cognitive investment in learning activities, which collectively influence academic

achievement and overall educational outcomes, as supported by Melissa Bond et al. (2020). Academic engagement is operationally defined in this study as the measurable level of involvement of TVL cookery students as assessed through a structured questionnaire covering the indicators of interests, study habits, academic ability, teacher-related aspect, and on-instructional material, where higher composite scores indicate greater academic engagement.

In an international context, particularly in China, Liu et al. (2025) reported that academic engagement had a direct but relatively weak positive effect on academic performance, indicating that although higher engagement was associated with improved grades, the magnitude of this effect was modest. This suggests that while engagement contributes to learning outcomes, its impact may be influenced by other adaptation factors within the academic environment.

Additionally, in Lima, Peru, Cutipa-Flores et al. (2025) found that academic engagement was negatively related to academic procrastination, such that higher engagement corresponded with less delay in completing academic tasks. The study indicated that engagement significantly predicted the likelihood of avoiding procrastination, highlighting that active involvement in learning supports more proactive academic behaviors.

Moreover, in Beijing, China, Miao (2025) observed that higher levels of research self-efficacy were associated with increased academic engagement, and that heightened engagement contributed to improved achievement through sustained effort, persistence, and cognitive involvement. This result underscores

that engagement is a meaningful component of academic success, even when considering varying academic demands.

In a Philippine context, particularly in Zamboanga City, Yusop and Ajihil (2026) examined learners' academic engagement across cognitive, affective, behavioral, and social dimensions among Social Sciences students. The study revealed that students exhibited moderate to high levels of engagement overall, with particularly strong affective and cognitive involvement even though attitudes toward the subject varied. This suggests that, in this regional setting, students maintained consistent participation and emotional investment in learning, reflecting broad engagement despite differences in subject perception.

Meanwhile, in selected elementary schools in Cagayan de Oro City, Cedeño and Mongas (2025) found that pupils generally demonstrated high academic engagement, evident through active participation and sustained attention during classroom tasks. Nevertheless, certain instructional situations still produced lower engagement in some behavioral aspects, indicating that engagement levels can fluctuate within the national context depending on classroom environment and teaching practices.

Furthermore, a study in Negros Occidental by Amarilla et al. (2025) showed that senior high school learners maintained high academic engagement during the "new normal," despite disruptions to conventional schooling and the shift to remote learning. Findings revealed that engagement remained consistently strong across multiple domains and was relatively uniform across different demographic groups,

demonstrating that students' involvement in learning activities remained resilient even amid significant educational changes.

In a regional Philippine context, particularly in Davao Region, Baog et al. (2025) found that academic engagement was moderately extensive among students in City College of Davao, with results indicating that students who exhibited greater academic interest showed significantly higher engagement levels in mathematics learning activities. Specifically, academic interest accounted for 35.2 % of the variance in academic engagement, suggesting that students' intrinsic appreciation of the subject corresponded with how actively they participated and involved themselves in their learning process, thereby highlighting the role of interest in fostering sustained engagement.

*Interests.* Interests refer to a student's psychological state of focused attention, curiosity, and positive affect toward learning tasks, reflecting both situational interests triggered by specific activities and more enduring individual interest, which motivates sustained involvement in subject matter (Renninger & Hidi, 2016). Interests are operationally defined as the level of attention, curiosity, and active participation a student demonstrates in TVL track learning tasks, including their willingness to engage with concepts beyond minimum requirements, maintain enthusiasm for challenges, and invest effort in completing academic activities.

Firstly, in Nairobi, Kenya, Muthoni and Wanjiru (2023) reported that students' academic interest strongly predicted their engagement in secondary school science courses. Students who demonstrated higher levels of intrinsic

curiosity and enjoyment devoted more attention to tasks, participated actively in learning activities, and persisted longer in problem-solving exercises. This study highlighted that interest serves as a motivational factor that supports sustained involvement in academic work and promotes overall engagement.

Meanwhile, in Tokyo, Japan, Sato et al. (2022) found that situational interest significantly enhanced student engagement in online mathematics learning modules. When learning materials were designed to capture attention through novelty and relevance, students showed higher levels of participation, cognitive focus, and emotional investment in their coursework. The findings indicate that activating students' interest can increase both the intensity and duration of engagement in digital learning contexts.

Furthermore, in Berlin, Germany, Keller and Becker (2021) demonstrated that individual interest in academic topics contributed to sustained engagement across multiple learning dimensions. Their study showed that students with higher personal interest engaged more frequently in self-directed learning, sought additional resources beyond classroom requirements, and maintained concentration during challenging tasks. These results underscore the role of interest as a critical driver of engagement, particularly in promoting long-term academic involvement.

Initially, in Negros Occidental, Bearneza (2025) found that students' *self-interest in mathematics* corresponded with varying levels of learning engagement, with male learners exhibiting significantly higher engagement scores in attentiveness, effort, and persistence in mathematical tasks compared to female

learners. This result suggests that stronger personal interest in mathematics can be linked to more active involvement and sustained effort in academic activities, highlighting the role of interest in shaping how students engage in challenging subject areas.

Meanwhile, in Cagayan de Oro City, Caay and Valle (2025) reported that learners' positive interest in academic subjects was strongly correlated with academic engagement within the school environment. Students who showed interest in their coursework also tended to exhibit higher levels of social and academic engagement, indicating that learners' interest can foster broader involvement in classroom interactions and collaborative activities, which supports their continued participation in schooling processes.

Furthermore, in Cagayan de Oro schools, Pioquinto and Oco (2025) observed that pupils' inspiration for reading—a form of academic interest—was significantly associated with higher levels of learning engagement. The findings indicated that students who were more inspired by reading showed stronger engagement across learning tasks, suggesting that interest-driven activities can enhance learners' attention, curiosity, and sustained involvement in academic contexts.

In a local context, particularly in Davao Oriental, Bugwak et al. (2025) reported that students' academic and non-academic engagement was influenced by several learning factors, where learners who experienced active learning strategies and quality instructional experiences exhibited higher levels of engagement, suggesting that intrinsic interest and meaningful participation in

learning activities helped sustain their involvement in both academic and extracurricular domains. The study indicated that when students perceive learning environments and instructional approaches as engaging and relevant, they tend to participate more actively and show stronger emotional and cognitive investment in school tasks, reinforcing the role of interest-driven learning conditions in promoting overall academic engagement.

*Study habits.* Study habits refer to the regular practices and routines that students engage in to acquire, organize, and retain knowledge, including effective time management, consistent review of learning materials, and purposeful engagement in academic tasks (Khan et al., 2021). Study habits is operationally defined as the frequency, consistency, and quality of a student's learning behaviors in the TVL track, encompassing planned study schedules, organized note-taking, reviewing lessons, and persistence in completing tasks, which collectively indicate the student's ability to manage learning efficiently and maintain active academic engagement.

Initially, in Melbourne, Australia, a study by Kutina and Francis (2025) found that study habits such as regular study schedules, time management, and active review routines were positively associated with academic engagement among undergraduates. Students who practiced consistent study routines reported greater participation in class discussions, higher persistence in academic tasks, and increased cognitive involvement, highlighting the role of disciplined study practices in sustaining engagement in formal learning environments.

In addition, in London, United Kingdom, research by Smith et al. (2024) showed that structured study habits—such as scheduled revision times and effective note-taking—were linked with higher levels of academic engagement among university learners. The findings indicated that students with well-developed study routines were more attentive, persistent in assignments, and more actively engaged in academic activities than peers with irregular or ineffective study practices.

Moreover, in Ontario, Canada, research conducted by Lee and Gonzalez (2024) reported that study habits significantly predicted students' engagement in online course environments, revealing that learners who consistently organized their study time, reviewed learning materials regularly, and minimized distractions showed higher overall academic engagement in asynchronous learning contexts compared to those with poor study routines.

Meanwhile, in Quezon City, a national study by Bulado et al. (2025) found that study habits such as goal setting, planning, and consistent review were strongly correlated with academic engagement among junior high school students. The research pointed out that students who exhibited strong study behaviors showed higher engagement levels, including greater participation, concentration, and persistence in class tasks.

Additionally, in Iloilo City, research by Rivera and Pascual (2024) reported that students who adopted organized study routines demonstrated higher academic engagement, particularly in terms of attentiveness, persistence, and active involvement in classroom activities. These findings support the view that

disciplined study habits contribute meaningfully to how engaged students are in their academic work.

Furthermore, in Cebu City, a national study by Samar and Dizon (2023) indicated that study habits significantly influenced academic engagement among senior high school learners, with students who reported regular review schedules and structured study behaviors also reporting higher participation and motivation in their learning activities. This underscores the importance of routine study practices in fostering student engagement.

Finally, in Davao City, Cullantes et al. (2024) found that study habits such as time management and organized study practices were positively correlated with online academic engagement among university students. The study revealed that students with regular study patterns were more consistent in participating in online discussions, submitting assignments on time, and engaging with course content, demonstrating that disciplined study habits support engagement even in hybrid and virtual learning contexts.

*Academic ability*. Academic ability refers to a student's capacity to understand, analyze, and apply knowledge effectively in academic tasks, encompassing intellectual skills such as comprehension, reasoning, problem-solving, and the practical application of learned concepts (Hidayat et al., 2021). Academic ability is operationally defined as the level of proficiency a student demonstrates in completing TVL track–related learning activities, measured through performance in written exams, practical exercises, and problem-solving

tasks, reflecting the student's mastery of subject matter and capability to apply knowledge in academic contexts.

Primarily, in Oslo, Norway, Thiel (2025) reported that students who maintained systematic study routines, such as organized review schedules and active study techniques, displayed significantly higher academic engagement. The study highlighted that consistent study habits enhanced participation, attention, and persistence in classroom tasks, reinforcing the role of disciplined learning behaviors in sustaining engagement in higher education.

Similarly, in Riyadh, Saudi Arabia, Aljaffer (2024) found that structured study habits, including regular study hours and effective self-monitoring, were strongly associated with academic engagement among medical students. The findings emphasized that learners with organized study routines were more focused, persistent, and active in academic tasks, suggesting the critical role of study habits in maintaining engagement.

Concurrently, in Islamabad, Pakistan, Lee and Gonzalez (2024) demonstrated that specific study habit practices, such as scheduling study sessions and using active learning strategies, correlated positively with academic engagement in online learning contexts. Students who maintained these habits showed increased participation and attention to learning materials, indicating that even in digital environments, disciplined study practices support engagement.

Notably, in Quezon City, Philippines, Bulado et al. (2025) reported that students' study habits, measured through goal setting, time management, and consistent study routines, were significantly linked to higher academic

engagement in junior high school classrooms. Learners who practiced disciplined study habits demonstrated increased participation, focus, and persistence in academic tasks, reinforcing the importance of effective study routines for engagement.

Equally, in Iloilo City, Philippines, Panday and Napil (2024) found that organized study routines, such as frequent review and structured study schedules, positively influenced students' engagement in classroom learning activities. Learners with well-maintained study habits exhibited higher attentiveness and active involvement, indicating that structured study behavior strengthens academic engagement.

Correspondingly, in Cebu City, Philippines, Samar and Dizon (2023) demonstrated that students with effective study habits—such as consistent preparation and self-regulated study techniques—exhibited greater academic engagement. These learners were more attentive, persistent, and actively involved in completing learning tasks, underscoring the link between disciplined study habits and engagement.

In a local context, particularly in Nabunturan, Davao de Oro, Philippines, Paynandos and Doronio (2025) found that study habits and student engagement were generally low among, indicating that learners' inconsistent study behaviors corresponded with lower levels of academic involvement in classroom tasks. The findings suggest that when students' study habits such as time allocation for review, homework routines, and preparation practices are not consistently developed, their engagement in academic activities may be diminished, pointing

to the need for interventions that strengthen habitual study routines to foster sustained academic engagement.

*Teacher related aspect.* Teacher Related Aspect refers to the influence of teachers' behaviors, instructional strategies, and support on students' learning experiences and academic engagement. It encompasses factors such as teacher guidance, feedback, clarity of instruction, motivation, and responsiveness to student needs (Moussa, 2021). Teacher Related Aspect is operationally defined as the extent to which TVL cookery students perceive their teachers' support, clarity in explaining lessons, provision of feedback, and motivational behaviors, measured through student survey responses indicating the frequency and quality of teacher interactions that enhance participation, understanding, and involvement in classroom activities.

Primarily, in China, Wang et al. (2021) reported that perceived teacher autonomy support significantly predicted higher academic engagement among secondary school students. Using structural equation modeling, the study found that teacher-related support had a direct positive effect on behavioral and emotional engagement, with standardized coefficients exceeding $\beta = .40$. Students who perceived their teachers as supportive, responsive, and autonomy-enhancing demonstrated stronger classroom participation and persistence in academic tasks, reinforcing the importance of teacher-related factors in sustaining engagement.

Similarly, in Spain, García-Moya et al. (2020) found that teacher support was strongly associated with adolescents' school engagement. Multilevel analyses revealed that perceived teacher support significantly predicted engagement,

explaining a meaningful proportion of variance in students' behavioral involvement. Learners who reported consistent instructional guidance and positive teacher interactions exhibited higher attentiveness and active participation in school activities.

Concurrently, in the United States, Quin et al. (2020) demonstrated that positive teacher–student relationships significantly predicted academic engagement across grade levels. Meta-analytic findings showed statistically significant effect sizes linking teacher support with engagement outcomes, indicating that supportive and low-conflict teacher relationships were consistently associated with higher student involvement and persistence in academic work.

Notably, in Metro Manila, Philippines, Datu and Park (2021) reported that teacher autonomy support significantly influenced Filipino students' academic engagement. Structural equation modeling results indicated that teacher-related support variables had significant direct effects on engagement indicators, accounting for substantial variance in behavioral and emotional engagement. Students perceiving higher levels of teacher encouragement displayed stronger academic involvement and sustained effort.

Equally, in Central Luzon, Philippines, Bernardo et al. (2022) found that perceived teacher support positively predicted academic engagement among senior high school students. Regression analysis revealed that teacher instructional practices and feedback significantly contributed to engagement scores, with teacher support explaining over 20% of the variance in engagement

levels. Learners who experienced structured guidance and constructive feedback were more focused and participative in classroom tasks.

Correspondingly, in Cebu City, Philippines, Malolos et al. (2023) demonstrated that teacher-related classroom support significantly predicted students' academic engagement. Quantitative findings showed a significant positive correlation between teacher support and engagement ($r > .45$, $p < .001$), indicating that students who perceived their teachers as approachable and supportive were more attentive and persistent in learning activities.

In a local context, particularly in Davao Region, Philippines, Ramos et al. (2024) found that teacher-related instructional support significantly influenced academic engagement among public senior high school students. Multiple regression analysis revealed that teacher support accounted for a significant percentage of variance in engagement scores, suggesting that consistent instructional guidance and encouragement enhance students' active participation and persistence in academic tasks.

*On instructional materials*. On-instructional materials refer to the teacher-provided, curriculum-aligned resources such as textbooks, worksheets, visual aids, multimedia tools, and demonstration materials that facilitate students' understanding, practice, and application of learning content, thereby promoting active engagement in academic tasks (Afflerbach et al., 2018). In this study, on-instructional materials are operationally defined as the extent to which TVL Cookery students perceive the availability, clarity, relevance, and usefulness of the instructional resources provided by teachers during classroom and skills-based

instruction, measured through a standardized survey assessing students' evaluation of these materials.

Primarily, in Australia, Martin and Dowson (2020) found that the quality and alignment of instructional materials significantly predicted students' academic engagement in secondary classrooms. The study revealed that when teachers used well-structured, relevant, and grade-appropriate materials, students displayed higher behavioral and cognitive engagement, suggesting that instructional resources that are coherent with lesson goals support deeper participation and focused attention.

Similarly, in the United Kingdom, Oliver et al. (2021) reported that access to varied digital instructional materials—such as interactive presentations, online simulations, and multimedia assignments—was positively associated with university students' academic engagement during blended learning. Students who perceived instructional materials as accessible, interactive, and supportive exhibited higher levels of persistence and intellectual involvement in academic tasks.

Concurrently, in the United States, Cho and Afflerbach (2022) demonstrated that instructional materials designed with scaffolding features (e.g., guided prompts, worked examples) were correlated with increased engagement in middle school literacy programs. Learners who worked with materials that prompted self-regulation and reflection showed greater attention, effort, and time on task, indicating that instructional design quality contributes meaningfully to academic engagement.

Notably, in Metro Manila, Cruz et al. (2021) found that the adequacy of teacher-provided instructional materials significantly correlated with senior high school students' academic engagement. Results indicated that classrooms with higher ratings of textbook relevance, supplementary handouts, and laboratory aids had students reporting greater participation and sustained effort in learning activities.

Equally, in Central Visayas, Dela Rosa and Bernardo (2022) reported that instructional materials that integrated real-world contexts (e.g., case studies, practical examples) positively influenced learners' engagement in general education courses. Students exposed to contextualized learning resources demonstrated higher emotional and cognitive engagement compared to peers given generic materials, suggesting the influence of material relevance on engagement dimensions.

Correspondingly, in Northern Mindanao, Gonzales et al. (2023) observed that clear, organized, and adequate instructional materials were predictors of academic engagement among secondary school learners. The study's regression model showed that instructional material quality explained a significant portion of the variance in engagement scores, with students who rated materials as clear and useful demonstrating enhanced participation and persistence.

In a Davao Region context, particularly in Davao City, Santos and Villanueva (2024) found that the perceived usefulness and availability of teacher-provided instructional materials were positively associated with students' academic engagement in senior high school classrooms. The research reported

that when learners identified instructional resources—such as skill manuals, visual aids, and demonstration tools—as relevant and supportive, they exhibited higher behavioral engagement, such as task persistence and active involvement in class activities.

**Relationship Between Student Preferences on Technical-Vocational Livelihood Track and Academic Engagement**

The relationship between student preferences on the TVL track and academic engagement refers to how learners' motivations, choices, and perceived support within their selected technical-vocational-livelihood specialization influence their active participation, attention, and effort in academic tasks. Conceptually, when students exhibit strong preferences for their chosen TVL strand—shaped by personal interest, encouragement from teachers, peers, and community, perceived opportunities, and alignment with career goals—they are more likely to demonstrate higher behavioral, emotional, and cognitive engagement, including persistence, attentiveness, and effective use of instructional materials (Skinner et al., 2022). Operationally, this relationship is defined as the degree to which TVL Cookery students' preference indicators predict measurable engagement outcomes, assessed through survey scores on task participation, focus, classroom involvement, and utilization of teacher-provided instructional resources.

Primarily, in Mexico City, Bazan-Perkins and Santibañez-Salgado (2025) found that students' learning style preferences particularly kinesthetic and sensory-based choices were significantly associated with higher engagement

outcomes in medical education, with learners who preferred experiential, practical learning showing stronger behavioral and cognitive engagement. These results suggest that when instructional contexts align with students' preferred ways of learning, academic engagement increases, supporting the notion that preference dimensions can positively influence engagement in higher-order educational settings.

Similarly, in Portugal, Ferreira and Veiga (2025) observed that students who reported stronger preferences for contextually relevant and well-structured learning materials also demonstrated higher engagement scores across behavioral, emotional, and cognitive dimensions, indicating a positive link between learners' educational preferences and their active involvement in academic tasks. These findings reinforce that preference alignment enhances academic participation.

Concurrently, in Denmark, Nielsen et al. (2025) demonstrated through large-scale survey research that students who expressed clear preferences for curriculum relevance and practical learning opportunities reported significantly higher academic engagement, particularly in experiential and interactive learning environments. This relationship underscores that when students' preferences are acknowledged in curriculum design, their engagement in learning activities tends to be higher.

Notably, in Metro Manila, Vergara (2023) examined the relationship between senior high school students' preferences for the TVL track at Bagbag National High School and their educational engagement. Pearson correlation

analyses indicated a significant positive relationship ($r > .40$, $p < .01$), showing that stronger preferences for TVL career-related factors were associated with higher engagement levels, suggesting that when students choose TVL tracks based on personal and career interests, they tend to engage more actively in academic tasks.

Similarly, in Cebu City, Navarro et al. (2024) found that Filipino university students whose instructional experiences matched their learning preferences demonstrated higher academic engagement in virtual education settings. The study reported that perceived preference-fit predicted greater participation and persistence, suggesting that preference alignment is important for engagement outcomes in blended and online contexts.

Equally, Dela Rosa and Bernardo (2022) reported that contextualized instructional materials that reflected students' learning preferences were positively linked to academic engagement among higher education learners in the Philippines. Students exposed to materials aligned with their interests and preferences showed greater cognitive and emotional engagement, indicating that preference-matched resources bolster involvement in academic tasks.

In Davao City, Cabahug (2022) observed that senior high school students who had clear preferences for their chosen Technical-Vocational-Livelihood (TVL) track showed higher levels of academic engagement. Students who selected tracks based on personal interests, perceived career opportunities, and teacher encouragement were more attentive in class, more persistent in completing tasks, and more actively involved in skills laboratories. The findings suggest that strong

alignment between students' TVL preferences and their motivations is associated with greater involvement in both theoretical and practical learning activities.

**Theoretical Framework and Conceptual Framework**

Albert Bandura's (1977) Social Learning Theory explains that individuals develop behaviors, attitudes, and learning patterns through observation and interaction within their social environment. In relation to this study, the theory explains the relationship between students' preferences for the Technical Vocational Livelihood track and their academic engagement, as students form preferences and learning behaviors by observing parents, teachers, and peers who value vocational skills and practical learning. When cookery students observe positive reinforcement, successful role models, and supportive learning environments within the TVL track, they are more likely to develop favorable preferences toward the program and actively engage in academic and practical tasks. This process of social modeling strengthens both students' track preferences and their level of academic engagement, supporting the relationship examined in the study.

Eccles and Wigfield's (1983) Expectancy Value Theory explains that students' educational preferences and choices are influenced by their expectations of success and the value they attach to a learning task or pathway. In the context of this study, the theory supports the independent variable by explaining how students develop preferences for the Technical Vocational Livelihood track when they believe they can perform well in cookery subjects and when they perceive vocational education as useful for future employment and career goals. When

students value the TVL track and expect positive outcomes from it, they are more likely to choose and commit to the program, strengthening their preference for the track and guiding their educational decisions.

Fredricks, Blumenfeld, and Paris's (2004) Student Engagement Theory explains academic engagement as a multidimensional construct consisting of behavioral, emotional, and cognitive involvement in learning activities. In relation to the dependent variable of this study, the theory explains that academic engagement among TVL cookery students is demonstrated through active participation in class activities, sustained effort in completing academic and practical tasks, positive emotional responses toward learning, and the use of effective learning strategies. When students are engaged across these dimensions, they are more likely to persist in their studies and achieve positive learning outcomes, making the theory relevant in explaining academic engagement within vocational education settings.

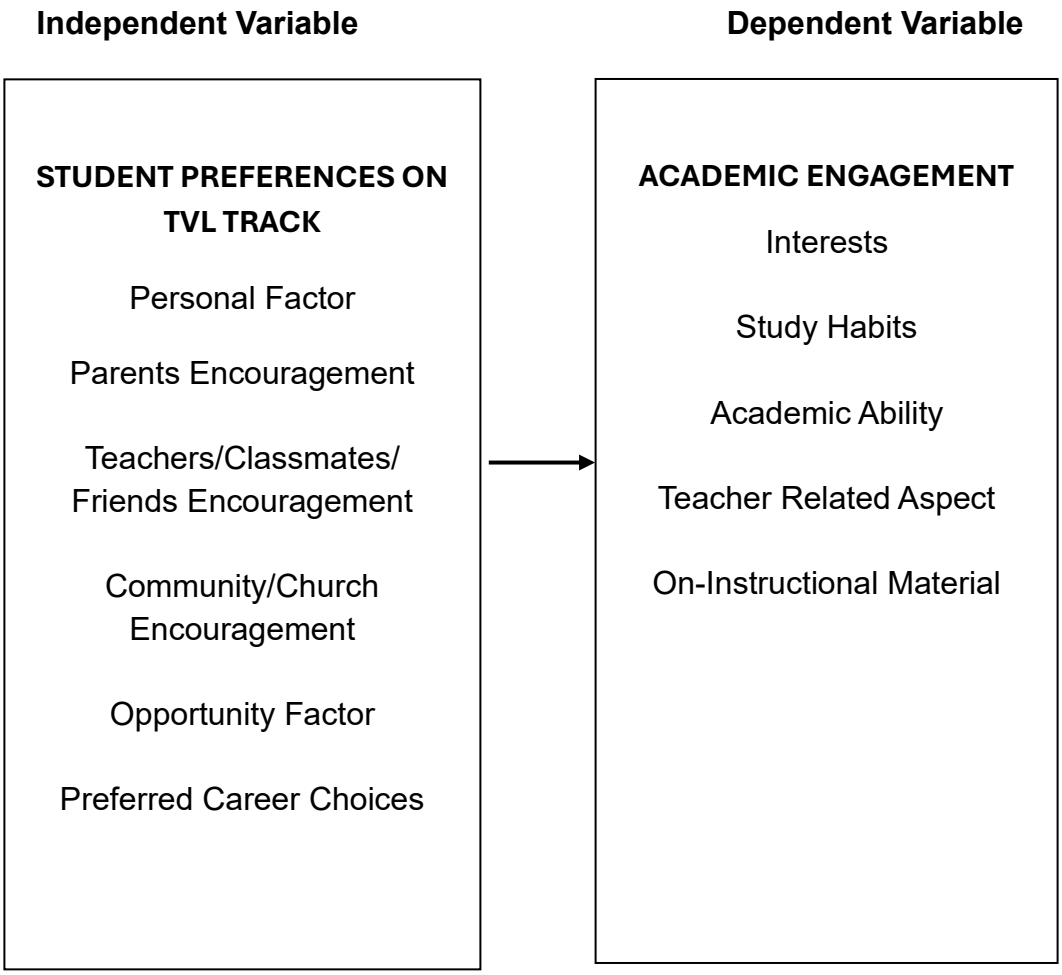


**Figure 1.** *Conceptual Framework Showing the Relationship between Student Preferences on TVL Track and Academic Engagement*

# CHAPTER 2

## Method

The methods and processes that was used to carry out this investigation are covered in this chapter. This section details the research design, respondents, ethical considerations, research instruments, data collection, and data analysis.

### Research Design

In this quantitative study design, the researchers used a non-experimental descriptive correlation method According to Miksza et al. (2023), descriptive correlational research design is a quantitative non-experimental method that seeks to describe variables and examine the relationships among them without manipulation. Researchers using this design typically employ surveys to collect data and then analyze the degree of association between variables as they naturally occur. The researchers consider that using a non-experimental descriptive correlation research method is suitable to address the research questions formulated, as it aims to investigate the Cookery Students' academic engagement.

### Research Respondents

The researchers used stratified random sampling as their sampling technique, where the population is divided into relevant subgroups (strata) and random samples are selected from each group to ensure proper representation; according to Saunders et al. (2019), this method increases accuracy and ensures

that each subgroup is adequately represented in the final sample. During the academic year, the school has a total of 1,233 students, with 244 students in the TVL strand. Since the respondents are limited to Cookery students, the researchers divided the Cookery strand into sections forming the strata, which include 41 Grade 11 students and 70 Grade 12 students. To determine the appropriate sample size, Slovin's Formula was used, resulting in 88 students selected from the TVL Cookery strand, with 33 questionnaires distributed to Grade 11 students and 55 questionnaires distributed to Grade 12 students. Because the samples from each stratum were chosen randomly, this helps minimize bias and allows the researchers to obtain a sample that accurately represents the studied population while ensuring fairness among respondents.

**Ethical Considerations**

The protection of human subjects by implementing appropriate ethical practices is crucial in any research study. In quantitative studies, ethical considerations (Yaw et al., 2023) have a specific resonance with respect to ensuring transparency, protecting respondents' rights, and maintaining integrity in the analysis of variables before the conduct of the said study. In accordance with Rizal Memorial Colleges Inc.'s Research Ethics Committee, the researchers made sure that the ethical considerations were rigorously followed and observed.

*Social Value.* As stated by Hitlin and Vaisey (2015), social values provide internalized standards that shape how individuals interpret experiences, evaluate their actions, and behave within their social environments. These values function as cognitive frameworks that influence expectations of acceptable conduct, guide

decision-making, and define the boundaries of desirable behavior in both personal and group contexts. By establishing shared norms of what is beneficial, appropriate, or meaningful, social values contribute to social coherence and inform collective expectations. This study has the primary goal of examining the relationship between student preferences on Technical-Vocational Livelihood Track and the Academic Engagement of Cookery students. With this, it will hold social value as it will help uncover significant insights into how Cookery students' preferences in the Technical-Vocational Livelihood Track relate to their level of academic engagement, thereby contributing to a deeper understanding of how values and choices influence learning behaviors in practical education settings.

*Informed Consent.* According to Dooly et al. (2017), informed consent is an essential ethical process that ensures respondents are fully aware of a study's purpose, procedures, and potential implications before agreeing to take part. It enables individuals to make voluntary and informed decisions about their participation, grounded in transparency, and respect for their autonomy. Before conducting the survey, the researchers will provide an informed consent form (ICF) to ensure that they will be fully aware of the basis on which the study will be conducted. Through this process, they will be informed of the study's purpose, procedures, potential risks (if any), anticipated benefits, and their rights and responsibilities as participants in this research endeavor.

*The Vulnerability of the Research Participant.* Identifying and addressing participant vulnerability is vital in ensuring ethical research, as it acknowledges that certain individuals may be more exposed to risk, misunderstanding, or undue

influence throughout the research process. Researchers must therefore implement safeguards that protect the well-being and dignity of these participants, particularly when dealing with populations who may experience limited decision-making capacity or increased sensitivity to personal information (Luna & VanderPoel, 2018). One of the vulnerabilities that will be identified among the research repondents will be the possibility of discomfort when disclosing information related to their academic confidence and learning habits. The respondents may become concerned that their responses will be shared with teachers, parents, or peers, even if confidentiality is assured. The researchers will address these vulnerabilities by ensuring strict confidentiality of responses, emphasizing the voluntary nature of participation, and creating a safe, non-judgmental environment for responses.

*Risks, Benefits, and Safety.* Participating in this research may offer students valuable insights into their learning behaviors and contribute to improved academic support systems. To ensure respondents' well-being, clear ethical guidelines are followed to prevent emotional discomfort, protect personal information, and avoid any form of harm. The primary objective is to advance educational knowledge while prioritizing the safety and welfare of all respondents, achieved through informed consent, confidentiality, and ethical safeguards (Bhandari, 2024). Conducting a study on Cookery students' preferences in the Technical-Vocational Livelihood Track and their academic engagement will involve minimal risks, such as mild discomfort when reflecting on personal learning habits or concerns about privacy when answering the survey. However, it will also provide benefits by increasing students' awareness of their study behaviors and offering insights that

may help improve instructional strategies and support systems in technical-vocational education. To ensure safety, the researchers will obtain informed consent, keep all responses anonymous and confidential, secure ethical approval, and guarantee that participation will remain voluntary.

*Privacy and Confidentiality of Information.* Safeguarding the privacy of research participants is a fundamental aspect of ethical research, as it protects individuals from unwanted exposure of their personal information. Ensuring confidentiality involves limiting access to participant data, preventing unauthorized disclosure, and implementing procedures that keep identities private throughout the research process. Respondents must be assured that their information will only be used for research purposes and stored securely to maintain their trust and dignity (Israel & Hay, 2020). This study aimed to adhere to the Data Privacy Act of 2012 or Republic Act 10173. This is to ensure that the said law is followed to the highest regard, wherein no names or information regarding the respondents are spread outside of the study ensuring absolute confidentiality.

*Justice.* This ethical principle emphasizes the fair and equitable treatment of research respondents, ensuring that the burdens and benefits of the study are distributed without discrimination. Researchers must protect vulnerable groups from exploitation, guaranteeing that no respondent is unfairly disadvantaged or excluded from potential benefits of the research. In applying justice, equal respect and consideration must be given to all participants, prioritizing their dignity and well-being throughout the research process (Beauchamp & Childress, 2019). The researchers will uphold impartiality by remaining unbiased and refraining from

giving any respondent preferential treatment, thereby ensuring absolute fairness. All respondents will be given sufficient time to complete the Google Form survey and will have the freedom to revise or withdraw any response they consider necessary or unsatisfactory. Since the survey will be administered online, respondents will ensure their own internet access using a device of their choosing.

*Transparency.* Transparency is a core ethical principle in research, ensuring that processes, decisions, and findings are openly communicated to respondents and stakeholders. When researchers clearly disclose study procedures, limitations, and intentions, they promote accountability and foster trust, enabling others to accurately interpret and evaluate the integrity of the research. Transparent practices also allow respondents to make well-informed decisions about their involvement, reinforcing respect, and ethical responsibility within the research community (Resnik, 2020). Before the conduct of the study, the researchers will provide a clear and sufficient explanation, including any necessary clarifications, regarding the methodology of the research. The procedures and processes will be thoroughly discussed to ensure that respondents fully understand what the study entails. This will allow them to decide freely whether they will choose to participate or withdraw from the study as they prefer.

Qualification of the Researcher. The qualification of the researcher involves possessing the necessary academic background, methodological competence, and strong ethical character essential for conducting rigorous research. It also includes the ability to apply appropriate techniques, exercise critical judgment, and maintain responsibility throughout the entire research process. These qualities

ensure that the researcher can uphold credibility, minimize errors, and strengthen the overall integrity of the study (Neuman, 2019). In this study, the help of a research adviser, panelists, and statistician will present to guide, correct, and guarantee the quality of the study. Their expertise will help address any shortcomings and will ensure that the study meets the standards required for scholarly and professional evaluation.

*Adequacy of Facilities.* The availability of complete, functional, and accessible facilities is essential in ensuring that research procedures are carried out accurately and efficiently. These facilities provide the necessary environment and tools that help minimize errors, maintain consistency, and support the proper execution of all research activities. Adequate and reliable resources also enhance the overall credibility and quality of the study by allowing researchers to perform tasks effectively (Kumar, 2019). Since the surveys will take place online, the researchers will ensure that the necessary digital resources, such as mobile phones, laptops, and stable internet connections, will be functional and accessible to the respondents. They will also verify that respondents have adequate internet access so that the data collection process will not be disrupted or rendered invalid.

*Community Involvement.* The involvement of the community plays a vital role in ensuring that the research is grounded in real experiences, needs, and perspectives of the people directly affected by the study. This participation strengthens the research process by promoting transparency, encouraging cooperation, and improving the accuracy and relevance of the data collected. Active engagement also enhances the credibility and social value of the research

by fostering trust, shared responsibility, and collective support (Israel et al., 2019). After the completion of this study, the researchers will provide a copy of the research to the research panel and to Rizal Memorial Colleges Inc. so that the findings will be reviewed, and any necessary revisions may be applied. In addition, a separate copy will be secured for future researchers who may wish to explore similar concepts and variables related to this study

**Research instruments**

To collect information for the independent variable, an adopted questionnaire was utilized to gather data on student preferences on Technical-Vocational Livelihood Track. The ranking listed below was applied to the respondents' answers to the questionnaire in order to determine the extent of Cookery students' preferences toward the Technical-Vocational Livelihood (TVL) track: 4 – Strongly Agree; 3 – Agree; 2 – Disagree; and 1 – Strongly Disagree.

Following that, the outcomes will be explained using the Likert scale below.

| **Range of means** | **Description** | **Interpretation** |
|---|---|---|
| 3.25 – 4.00 | Strongly Agree | This means that students' preferences toward the TVL track among Cookery students are highly encouraged. |
| 2.50-3.24 | Agree | This means that students' preferences toward the TVL track among Cookery students are encouraged. |

| | | |
|---|---|---|
| 1.75 -2.49 | Disagree | This means that students' preferences toward the TVL track among Cookery students are moderately encouraged. |
| 1.00- 1.74 | Strongly Disagree | This means that students' preferences toward the TVL track among Cookery students are not encouraged. |

To collect information for the dependent variable, an adopted questionnaire was utilized to gather data on students' academic engagement. The ranking listed below was applied to the respondents' answers to the questionnaire in order to determine the extent of academic engagement among Cookery students: 5 – Strongly Agree; 4 – Agree; 3 – Disagree; and 1 – Strongly Disagree.

Following that, the outcomes will be explained using the Likert scale below

| **Range of means** | **Description** | **Interpretation** |
|---|---|---|
| 3.25-4.00 | Strongly Agree | This means that the academic engagement of Cookery students is very positively engaged |
| 2.50-3.24 | Agree | This means that the academic engagement of Cookery students is positively engaged |
| 1.75 -2.49 | Disagree | This means that the academic engagement of Cookery students is negatively engaged |

| | | |
|---|---|---|
| 1.00- 1.74 | Strongly Disagree | This means that the academic engagement of Cookery students is Very negatively engaged |

The independent variable research instrument has a total of 30 questions, 5 for each indicator. Meanwhile, the dependent variable research instrument has 50 questions in total, with 10 questions per indicator. In summary, the respondents answered a total of 80 questions.

**Data Collection**

The following processes was utilized to collect the data required for this study.

***Requesting approval to carry out the research.*** After receiving approval from the research ethics committee and under the direction of the thesis adviser, the researchers will seek an endorsement letter from the principal of the Rizal Memorial Colleges, Inc. Tagum Campus to carry out the study to determine the relationship between Student preferences on Technical-Vocational Livelihood Track and Academic Engagement. The researchers will write a letter of request to the school principal, asking for permission to use the school as the research location. In adherence to ethical standards, informed consent will be obtained to ensure that all respondents voluntarily agree to take part in the study.

***Administration and questionnaire retrieval.*** With the principal's authorization, the researchers distributed the research instrument using Google Forms. To promote participation and ensure that all students receive the survey, the researchers briefly visited each Grade 11 and Grade 12 TVL Cookery

classroom to explain the study and provide the survey link. The online form included questions assessing students' preferences regarding the Technical-Vocational Livelihood Track and their level of academic engagement, beginning with an informed consent section that explains the study's goals and assures the confidentiality of all responses. Students will have two days to answer the questionnaire at their convenience, after which all responses will be automatically collected through Google Forms.

***Data collection and tabulation.*** All responses submitted through Google Forms was automatically recorded in a secure online spreadsheet. The researchers exported the dataset and perform tabulation to organize and prepare the data for statistical analysis. The data that was gathered and tabulated for both independent and dependent variables was tallied. Upon completion of the survey, the collected data was systematically organized and rendered to a qualified statistician for rigorous data analysis

**Data Analysis**

The study's findings was and explained using the following statistical tools.

***Mean.*** In statistics, the mean is a measure of central tendency calculated by summing all values in a dataset and dividing the total by the number of observations (MacGillivray, 2014). In this study, the mean was used to determine the overall level of students' preferences toward the Technical-Vocational Livelihood (TVL) Track and their level of academic engagement among Grade 11 and Grade 12 Cookery students.

***Pearson Product Moment Coefficient Correlation.*** The Pearson Product Moment Correlation Coefficient is a statistical method used to determine the strength and direction of the relationship between two continuous variables, indicating whether the association is weak or strong and whether it is positive or negative (Schober et al., 2018). In this study, Pearson r was employed to examine whether a significant relationship exists between students' preferences toward the Technical-Vocational Livelihood (TVL) Track and their level of academic engagement among Grade 11 and Grade 12 Cookery students.

# Chapter 3

## Results and Discussions

In this chapter, the researchers have figured out and interpreted the data that have been gathered from the respected respondents based on their perspectives and experiences. The discussions are organized according to the problem statement's placement and have determined the respondents' responses.

*The Level of Student Preferences on Technical-Vocational Livelihood Track*

The first statement of the problem is to determine the level of student preferences on the TVL track among Senior High School students. Table 1 presents the answer to this question. Student preferences on the TVL track consist of the following indicators: personal factor, parents' encouragement, teachers, classmates, and friends' encouragement, community or church encouragement, opportunity factor, and preferred career choices. The table also shows the mean and its corresponding descriptive value of each of these indicators, including the overall values.

*Table 1. Students Preferences on TVL Track*

| Indicators | Mean | Description |
|---|---|---|
| Personal Factors | 3.0265 | Agree |
| Parents Encouragement | 3.0152 | Agree |
| Teachers/Classmates/Friends Encouragement | 2.9795 | Agree |

| | | |
|---|---|---|
| Community and Church Encouragement | 2.9602 | Agree |
| Opportunity Factor | 3.0814 | Agree |
| Preferred Career Choice | 2.5777 | Agree |
| **OVERALL** | 2.9401 | Agree |

The overall value of the mean that determines the level Student Preferences on TVL Track of the Cookery students in Rizal Memorial Colleges, Inc. is equal to 2.9401, which corresponds to the descriptive value of "Agree." Based on the result, the Student Preferences on TVL Track of the cookery students at Rizal Memorial Colleges, Inc., is classified and interpreted that students' preferences toward the TVL track among Cookery students are encouraged.

As shown in Table 1, among the indicators, "Opportunity Factor" obtained the highest mean (3.0814), suggesting that Opportunity factors are particularly helpful in strengthening students' preferences for the TVL track through the availability of accessible learning resources and opportunities that support skill development and understanding.

As shown in Table 1, among the indicators, "Personal Factor" obtained the second highest mean (3.0265), suggesting that personal factors are particularly helpful in strengthening students' preferences for the TVL track through their individual interests, motivations, and self-related considerations that influence their academic choices.

As shown in Table 1, among the indicators, “Preferred Career Choices” obtained the lowest mean (2.5777), suggesting that preferred career choices are still helpful in strengthening students’ preferences for the TVL track, as respondents generally agree that their future career intentions influence their academic decisions, although to a lesser

*The Level of Student Preferences on Technical-Vocational Livelihood Track in terms of Personal Factors*

The following Table 1.1 indicates the level of student preferences on TVL track of cookery students in terms of personal factors. There are 6 statements in it, each having a meaning and description that corresponds to it.

*Table 1.1 The Level of Student Preferences on TVL Track in terms of Personal Factors*

| Personal Factors | Mean | Description |
|---|---|---|
| 1. I have participated in orientations about career choice, and TVL became my interest. | 3.3409 | Strongly Agree |
| 2. I have definitely made a choice to be in TVL track. | 3.2273 | Agree |
| 3. I have no career plans at that time. | 2.7500 | Agree |
| 4. I have done career research on my own and I found TVL as an answer to that. | 3.0909 | Agree |
| 5. I have considered applying a job after graduating to TVL. | 3.0682 | Agree |

| | | |
|---|---|---|
| 6. I was limited to choosing TVL because of my grades. | 2.6818 | Agree |
| **OVERALL** | 3.0265 | Agree |

As shown in Table 1.1, the statement with the highest mean of 3.3409 is “I have participated in orientations about career choice, and TVL became my interest.” It shows that the level of students’ preference on the TVL track in terms of personal factors is high and often present. This reflects a result that students’ participation in career-related orientations strongly influences their interest in choosing the TVL track.

It is also evident in Table 1 that the statement, “I was limited to choosing TVL because of my grades,” has the lowest mean of 2.6818, which is described as agree and is sometimes present. The result reflects a conclusion that students’ preference for the TVL track is somewhat influenced by their academic grades, although this factor has less impact compared to other personal factors.

*The Level of Student Preferences on Technical-Vocational Livelihood Track in terms of Parents Encouragement*

The following Table 1.2 indicates The Level of Student Prefereneces on TVL Track in terms of Parents Encouragement. There are 3 statements in it, each having a meaning and description that corresponds to it.

*Table 1.2 The Level of Student Prefereneces on TVL Track in terms of Parents Encouragement*

| Parents Encouragement | Mean | Description |
|---|---|---|
| 1. My parents were supportive on my decisions upon choosing TVL. | 3.1136 | Agree |
| 2. My parents kept on motivating me to pursue TVL. | 3.0114 | Agree |
| 3. My parents encouraged me to pursue in the TVL track | 2.9205 | Agree |
| **OVERALL** | 3.0152 | Agree |

As shown in Table 1.2, the statement with the highest mean of 3.1136 is “My parents were supportive on my decisions upon choosing TVL.” It shows that the level of students’ preference on the TVL track in terms of parents’ encouragement is encouraged and often present. This reflects a result that parental support plays an important role in strengthening students’ interest and commitment to the TVL track.

It is also evident in Table 1.2 that the statement, “My parents encouraged me to pursue in the TVL track,” has the lowest mean of 2.9205, which is described as agree and is sometimes present. The result reflects a conclusion that parental encouragement still influences students’ preference for the TVL track, although its impact is slightly less compared to other statements under the same indicator.

*The Level of Student Preferences on Technical-Vocational Livelihood Track in terms of Teachers/Classmates/Friends Encouragement*

The following Table 1.3 indicates the level of The Level of Student Preferences on TVL Track in terms of Teachers/Classmates/Friends Encouragement There are 5 statements in it, each having a meaning and description that corresponds to it.

*Table 1.3. The Level of Student Preferences on TVL Track in terms of Teachers/Classmates/Friends Encouragement*

| Teachers/Classmates/Friends Encouragement | Mean | Description |
|---|---|---|
| 1. My teachers involved me in career-related TVL activities in the classroom. | 3.0909 | Agree |
| 2. My teachers were talking to me about TVL. | 3.0341 | Agree |
| 3. My friends motivated me in choosing TVL which is suitable for my skills. | 2.9659 | Agree |
| 4. My friends helped me in choosing TVL which is suitable for my skills | 2.9091 | Agree |
| 5. I want to follow where majority of my classmates were choosing TVL. | 2.8977 | Agree |
| **OVERALL** | 2.9795 | Agree |

As shown in Table 1.3, the statement with the highest mean of 3.0909 is “My teachers involved me in career-related TVL activities in the classroom.” It shows that the level of students’ preference on the TVL track in terms of teachers/classmates/friends encouragement is encouraged and often present. This reflects a result that involvement of teachers in career-related activities positively influences students’ interest and engagement in the TVL track.

It is also evident in Table 1.3 that the statement, “I want to follow where majority of my classmates were choosing TVL,” has the lowest mean of 2.8977, which is described as agree and is sometimes present. The result reflects a conclusion that classmates’ influence still affects students’ preference for the TVL track, although its impact is slightly less compared to other statements under the same indicator.

*The Level of Student Preferences on*
*Technical-Vocational Livelihood Track*
*in terms of Community and Church Encouragement*

The following Table 1.4 indicates the The Level of Student Preferences on TVL Track in terms of Community and Church Encouragement There are 4 statements in it, each having a meaning and description that corresponds to it.

*Table 1.4. The Level of Student Preferences on TVL Track in terms of Community and Church Encouragement*

| Community and Church Encouragement | Mean | Description |
|---|---|---|
| 1. My community organization gave | 3.1136 | Agree |

| | | |
|---|---|---|
| me the impression that TVL can make my life better. | | |
| 2. My community organization encouraged me to pursue TVL. | 3.0909 | Agree |
| 3. My church organization motivated me to choose TVL as my track. | 2.8182 | Agree |
| 4. My church organization helped me decide in choosing TVL as the appropriate track. | 2.8182 | Agree |
| **OVERALL** | 2.9602 | Agree |

As shown in Table 1.4, the statement with the highest mean of 3.1136 is “My community organization gave me the impression that TVL can make my life better.” It shows that the level of students’ preference on the TVL track in terms of community and church encouragement is encouraged and often present. This reflects a result that support from community organizations positively influences students’ interest and motivation in pursuing the TVL track.

It is also evident in Table 1.4 that the statement, “My church organization motivated me to choose TVL as my track,” has the lowest mean of 2.8182, which is described as agree and is sometimes present. The result reflects a conclusion that church organizations still play a role in encouraging students’ preference for the TVL track, although their influence is slightly less compared to other statements under the same indicator.

*The Level of Student Preferences on Technical-Vocational Livelihood Track in terms of Opportunity Factor*

The following Table 1.5 indicates The Level of Student Preferences TVL Track in terms Opportunity Factor. There are 8 statements in it, each having a meaning and description that corresponds to it.

*Table 1.5. The Level of Student Preferences TVL Track in terms Opportunity Factor*

| Opportunity Factor | Mean | Description |
|---|---|---|
| 1. The location of the school which offers TVL is accessible to my residence. | 3.1023 | Agree |
| 2. The TVL track is available in our community. | 3.1364 | Agree |
| 3. There are numerous job opportunities for TVL in our 4. community. | 3.0909 | Agree |
| 5. I have access to employment in a family business which 6. is related to TVL after I graduate. | 3.0455 | Agree |
| 7. TVL jobs are in demand in our community. | 3.0341 | Agree |
| 8. Money has been an issue that was why I choose TVL. | 3.0795 | Agree |

| | | |
|---|---|---|
| **OVERALL** | 3.0814 | Agree |

As shown in Table 1.5, the statement with the highest mean of 3.1364 is “The TVL track is available in our community.” It shows that the level of students’ preference on the TVL track in terms of opportunity factor is encouraged and often present. This reflects a result that the availability of the TVL track in the community provides students with accessible opportunities that support their interest and participation in the track.

It is also evident in Table 1.5 that the statement, “TVL jobs are in demand in our community,” has the lowest mean of 3.0341, which is described as agree and is sometimes present. The result reflects a conclusion that the demand for TVL-related jobs in the community still encourages students’ preference for the TVL track, although its influence is slightly less compared to other statements under the same indicator.

*The Level of Student Preferences on Technical-Vocational Livelihood Track in terms of Preferred Career Choice*

The following Table 1.6 indicates The Level of Student Preferences on TVL Track in terms Preferred Career Choice. There are 6 statements in it, each having a meaning and description that corresponds to it.

*Table 1.6. The Level of Student Preferences on TVL Track in terms Preferred Career Choice*

| Preferred Career Choice | Mean | Description |
|---|---|---|
| 1. I am satisfied that I graduated in the TVL track. | 2.5227 | Agree |
| 2. I am disappointed whenever I miss one activity. | 3.1705 | Agree |
| 3. I want to get good grades, on test quizzes, assignments and projects. | 2.4545 | Disagree |
| 4. I listen attentively to the lecture of my teacher. | 2.4773 | Disagree |
| 5. I actively participate in the discussion, doing activities and/ or clarifying things I did not understand. | 2.3409 | Disagree |
| 6. I am excited in engaging myself in the class that is related to TVL | 2.5000 | Agree |
| **OVERALL** | 2.5777 | Agree |

As shown in Table 1.6, the statement with the highest mean of 3.1705 is “I am disappointed whenever I miss one activity.” It shows that the level of students’ preference on the TVL track in terms of preferred career choice is encouraged and often present. This reflects a result that students’ interest in their future careers motivates them to actively participate in TVL activities, reinforcing their commitment to the track.

It is also evident in Table 1.6 that the statement, “I actively participate in the discussion, doing activities and/or clarifying things I did not understand,” has the

lowest mean of 2.3409, which is described as disagree and is sometimes present. The result reflects a conclusion that students' preferred career choices have a moderate influence on their active participation in TVL-related activities, showing that this factor is less encouraging compared to other statements under the same indicator.

*The Level of Student Preferences on Technical-Vocational Livelihood Track*

The second statement of the problem is to determine the level of academic engagement among Senior High School TVL Cookery students. Table 2 presents the answer to this question. Academic engagement consists of the following indicators: interest, study habits, academic ability, teacher-related aspect, and on-instructional materials. The table also shows the mean and its corresponding descriptive value of each of these indicators, including the overall values.

*Table 2. Academic Engagement of TVL cookery students*

| Indicators | Mean | Description |
|---|---|---|
| Interest | 3.2323 | Agree |
| Study Habits | 3.1489 | Agree |
| Academic Ability | 3.2262 | Agree |
| Teacher-Related Aspect | 3.2035 | Agree |
| On-Instructional Materials | 3.2459 | Agree |
| **OVERALL** | 3.2114 | Agree |

The overall value of the mean that determines the level of Academic Engagement of the Cookery students in Rizal Memorial Colleges, Inc. is equal to 3.2114, which corresponds to the descriptive value of "Agree." Based on the result, the Academic Engagement of the Cookery students at Rizal Memorial Colleges, Inc. is classified and interpreted that students' engagement toward their academic activities in the TVL track is encouraged. As shown in Table 1, among the indicators, "Opportunity Factor" obtained the highest mean (3.0814), suggesting that Opportunity factors are particularly helpful in strengthening students' preferences for the TVL track through the availability of accessible learning resources and opportunities that support skill development and understanding.

This result is supported by Baytos and Strang (2018), who found that TVL students' engagement in academic activities increases when they are provided with relevant instructional experiences and active learning opportunities that support their involvement. Their study shows that students participating in meaningful, context-related tasks are more likely to demonstrate consistent engagement, reflecting that supportive learning environments foster positive academic involvement. The similarity between the present findings and the cited study suggests that when TVL Cookery students are given accessible learning materials, structured opportunities, and engaging instructional support, their academic engagement is encouraged. Overall, the result implies that students' engagement toward their academic activities in the TVL track is sustained and positively influenced by the conditions and opportunities present in their learning context.

As shown in Table 2, the statement with the highest mean of 3.2459 is “On-Instructional Materials.” It shows that the level of Academic Engagement of the Cookery students in terms of on-instructional materials is encouraged and often present. This reflects a result that students actively use and engage with instructional materials to support their learning in the TVL track.

Research supports the finding that instructional materials play a significant role in encouraging academic engagement among vocational learners. For instance, Kartal (2020) found that the availability and quality of instructional resources in vocational education significantly enhanced students’ engagement and practical skill development, as learners who had access to diverse and relevant learning materials demonstrated higher levels of involvement in classroom and hands-on activities. The study also emphasized that when instructional materials are aligned with students’ learning needs and vocational tasks, students show greater participation and sustained interest in learning. The similarity between the present findings and the cited study suggests that TVL Cookery students actively use and engage with on-instructional materials to support their learning processes. Overall, the result implies that the presence and effective use of instructional materials encourage and sustain students’ academic engagement in the TVL track.

It is also evident in Table 2 that the statement, “Interest,” has the second highest mean of 3.2323, which is described as agree and is sometimes present. The result reflects a conclusion that students’ interest in their subjects encourages their Academic Engagement in the TVL track.

Research supports the finding that students' interest in learning significantly encourages academic engagement. In a study by Reeve et al. (2018), student interest was shown to be a strong predictor of engagement in classroom activities, where learners who reported higher levels of interest also demonstrated increased attention, participation, and persistence in their academic tasks. The study emphasized that when instructional content and tasks appeal to students' personal interests, engagement behaviors are more likely to occur and be sustained. The similarity between the present findings and the cited study suggests that TVL Cookery students' genuine interest in their subjects encourages their active involvement in academic activities. Overall, the result implies that students' interest plays a positive and influential role in supporting their academic engagement in the TVL track.

It is also evident in Table 2 that the statement, "Study Habits," has the lowest mean of 3.1489, which is described as agree and is sometimes present. The result reflects a conclusion that students' study habits still encourage their Academic Engagement in the TVL track, although its influence is slightly less compared to other indicators.

Research supports the finding that students' study habits contribute to academic engagement, even when their influence is moderate. In a study by Richardson et al. (2019), it was found that effective study habits such as regular review, time management, and self-regulated learning strategies were positively associated with students' engagement and persistence in academic tasks. The authors emphasized that while study habits may not be the strongest predictor of

engagement compared to other motivational factors, they nonetheless play a meaningful role in supporting sustained involvement in learning activities. The similarity between the present findings and the cited study suggests that TVL Cookery students' structured study habits continue to encourage their active participation in academic work. Overall, the result implies that consistent and organized study practices help promote academic engagement in the TVL track.

*The Level of Academic Engagement of Cookery Students in terms of Interests.*

The following Table 2.1 indicates The Level of Academic Engagement of Cookery Students in terms of Interests.. There are 9 statements in it, each having a meaning and description that corresponds to it.

*Table 2.1. The Level of Academic Engagement of Cookery Students in terms of Interests.*

| Interest | Mean | Description |
|---|---|---|
| 1. I am satisfied that I graduated in the TVL track. | 3.3295 | Strongly Agree |
| 2. I am disappointed whenever I miss one activity. | 3.1591 | Agree |
| 3. I want to get good grades, on test quizzes, assignments and projects. | 3.2955 | Strongly Agree |
| 4. I listen attentively to the lecture of my teacher. | 3.2841 | Strongly Agree |
| 5. I actively participate in the discussion, doing activities and/or clarifying | 3.2727 | Strongly Agree |

| | | |
|---|---|---|
| things I did not understand. | | |
| 6. I actively participate in the discussion, doing activities and/or clarifying things I did not understand. | 3.2727 | Strongly Agree |
| 7. I am excited in engaging myself in the class that is related to TVL. | 3.1705 | Agree |
| 8. I always make myself prepared for the subject. | 3.1932 | Agree |
| 9. I get frustrated when the discussion is interrupted or the teacher is absent. | 3.1136 | Agree |
| **OVERALL** | 3.2323 | Agree |

As shown in Table 2.1, the statement with the highest mean of 3.1705 is “I am disappointed whenever I miss one activity.” It shows that the level of students’ preference on the TVL track in terms of preferred career choice is encouraged and often present. This reflects a result that students’ interest in their future careers motivates them to actively participate in TVL activities, reinforcing their commitment to the track.

Research supports the finding that students’ career interests positively influence their engagement in academic and practical activities. For example, Fouad et al. (2016) reported that students who have clear career goals demonstrate higher motivation, persistence, and active participation in related learning tasks. The study emphasized that when students perceive a strong

connection between their academic activities and future career aspirations, their involvement and commitment increase. The similarity between the present findings and the cited study suggests that TVL Cookery students' interest in their future careers motivates them to actively participate in TVL activities. Overall, the result implies that students' preferred career choices encourage and sustain their engagement in the TVL track.

It is also evident in Table 2.1 that the statement, "I actively participate in the discussion, doing activities and/or clarifying things I did not understand," has the lowest mean of 2.3409, which is described as disagree and is sometimes present. The result reflects a conclusion that students' preferred career choices have a moderate influence on their active participation in TVL-related activities, showing that this factor is less encouraging compared to other statements under the same indicator.

Research indicates that students' career preferences can moderately affect their engagement in learning activities. Lent et al. (2018) found that students with clear career goals often show higher motivation and participation; however, the effect on their day-to-day engagement depends on additional factors such as self-efficacy and interest. This suggests that while career aspirations contribute to involvement, they may not be the primary driver of active participation. In line with this, the present study shows that TVL Cookery students' preferred career choices moderately influence their engagement in TVL-related activities. Overall, the result implies that students' career intentions support their participation, though the impact is less pronounced compared to other motivating factors in the TVL track.

*The Level of Academic Engagement of Cookery Students in terms of Study Habits*

The following Table 2.2 indicates The Level of Academic Engagement of Cookery Students in terms of Study Habits. There are 10 statements in it, each having a meaning and description that corresponds to it.

*Table 2.2. The Level of Academic Engagement of Cookery Students in terms of Study Habits*

| Study Habits | Mean | Description |
|---|---|---|
| 1. I have a specific place of study at home which I keep clean and orderly manner. | 3.2386 | Agree |
| 2. I exert more effort when I do difficult things. | 3.1250 | Agree |
| 3. I do my assignments regularly. | 3.1023 | Agree |
| 4. I do my assignments regularly. | 3.1705 | Agree |
| 5. I prefer finishing my studying and my assignments first before watching any television program. | 3.1136 | Agree |
| 6. I see to it that extracurricular activities do not hamper my studies. | 3.0682 | Agree |
| 7. I study the lessons I missed if I was absent from the class. | 3.1477 | Agree |
| 8. I spend my vacant time in doing assignments or studying my lessons. | 3.1818 | Agree |

| | | |
|---|---|---|
| 9. I study harder to improve my performance when I get low grades. | 3.1250 | Agree |
| 10. I spend less time with my friends during school days to concentrate more on my studies. | 3.2159 | Agree |
| **OVERALL** | 3.1489 | Agree |

As shown in Table 2.2, the statement with the highest mean of 3.2386 is “I have a specific place of study at home which I keep clean and orderly manner.” It shows that the level of Academic Engagement of the Cookery students in terms of study habits is encouraged and often present. This reflects a result that maintaining an organized study environment supports students’ focus and engagement in their TVL track activities.

This is supported by Assunção et al. (2025), who found that a well-defined and orderly learning space is a critical determinant of behavioral and cognitive engagement among high school students. Their research indicates that students who utilize a specific, dedicated study area at home exhibit higher levels of "vigor and absorption"—two key dimensions of academic engagement that allow them to sustain focus on complex, track-specific tasks. By minimizing environmental "noise" and physical clutter, students are better able to self-regulate their learning habits, which is particularly vital for Technical-Vocational-Livelihood (TVL) learners who must balance theoretical knowledge with the discipline required for practical applications.

It is also evident in Table 2.2 that the statement, “I see to it that extracurricular activities do not hamper my studies,” has the lowest mean of 3.0682, which is described as agree and is sometimes present. The result reflects a conclusion that students’ management of extracurricular activities still encourages their study habits, although its influence is slightly less compared to other statements under the same indicator.

This is supported by Anis and Shofawati (2024), who found that while extracurricular participation can create time constraints, it often serves as a "dual-edged sword" that can actually enhance academic engagement if managed correctly. Their research indicates that students who consciously limit the interference of outside activities with their core curriculum—such as those in specialized TVL tracks—demonstrate higher levels of "academic grit" and self-discipline. Although these students may experience a slight decrease in study time, the soft skills gained through extracurriculars, such as time management and social coordination, often provide a compensatory effect that keeps their overall study habits within a positive and productive range.

*The Level of Academic Engagement of Cookery Students in terms of Academic Ability*

The following Table 2.3 indicates The Level of Academic Engagement of Cookery Students in terms of Academic Ability. There are 11 statements in it, each having a meaning and description that corresponds to it.

*Table 2.3. The Level of Academic Engagement of Cookery Students in terms of Academic Ability*

| Academic Ability | Mean | Description |
|---|---|---|
| 1. I think this activity will be useful for the development of my future projects. | 3.2614 | Strongly Agree |
| 2. I think that this activity is relevant to the course and to my curriculum. | 3.2273 | Agree |
| 3. I prepare all the materials to be needed for our activity. | 3.2614 | Strongly Agree |
| 4. I consider that the development of this activity challenges my ability to perform. | 3.2500 | Strongly Agree |
| 5. I always get excited for our laboratory activity. | 3.2045 | Agree |
| 6. I think these activities will help my performance increase. | 3.2727 | Strongly Agree |
| 7. I receive help in problems related to the activity. | 3.1364 | Agree |
| 8. I think that this type of activities encourages me to work better. | 3.2614 | Strongly Agree |
| 9. I am satisfied with my opportunities to use tools and equipment. | 3.2159 | Agree |
| 10. I can say that my knowledge in TVL increased. | 3.2386 | Agree |
| 11. I have made progress in my ability in this course. | 3.1591 | Agree |
| **OVERALL** | 3.2262 | Agree |

As shown in Table 2.3, the statement with the highest mean of 3.2727 is “I think these activities will help my performance increase.” It shows that the level of Academic Engagement of the Cookery students in terms of academic ability is highly encouraged and often present. This reflects a result that students recognize the value of TVL track activities in improving their academic performance.

This is supported by Zaccoletti et al. (2025), who found that students' academic engagement is significantly intensified when they perceive a direct link between their current school activities and their future competency levels. Their research suggests that when learners—particularly those in vocational or practical tracks—believe that specific tasks will lead to a tangible increase in performance, they develop a higher sense of "academic self-efficacy." This cognitive appraisal acts as a motivational catalyst, encouraging students to invest more effort and persist through the rigorous requirements of their specialized curriculum, as they view these activities not as mere requirements but as essential tools for their professional development.

It is also evident in Table 2.3 that the statement, “I receive help in problems related to the activity,” has the lowest mean of 3.1364, which is described as agree and is sometimes present. The result reflects a conclusion that receiving assistance in activity-related problems still encourages students’ academic ability, although its influence is slightly less compared to other statements under the same indicator.

This is supported by Li et al. (2023), whose study on higher vocational students revealed that while external assistance and instructional support are

critical for academic engagement, their direct impact is often perceived as a secondary driver compared to a student's internal learning drive. Their research indicates that receiving help during complex practical activities provides essential "competency support," which mitigates student frustration and helps them navigate difficult hands-on tasks. However, because this scaffolding serves primarily as an external facilitator rather than an intrinsic motivator, its overall influence is frequently rated slightly lower by students than their own self-efficacy or personal belief in the task's value. Nonetheless, this assistance remains a crucial supportive element that encourages and sustains their overall academic ability.

*The Level of Academic Engagement of Cookery Students in terms of Teacher-Related Aspect*

The following Table 1.2 indicates The Level of Academic Engagement of Cookery Students in terms of Teacher-Related Aspect. There are 11 statements in it, each having a meaning and description that corresponds to it.

*Table 2.4 The Level of Academic Engagement of Cookery Students in terms of Teacher-Related Aspect*

| Teacher-Related Aspect | Mean | Description |
|---|---|---|
| 1. Guides the student in performing the activity. | 3.2727 | Strongly Agree |
| 2. Gives examples and ideas. | 3.1818 | Agree |
| 3. Explains the objectives of the lesson clearly at the start of each period. | 3.2159 | Agree |

| | | |
|---|---|---|
| 4. Activities planned to augment information on TVL. | 3.1818 | Agree |
| 5. Is updated with present trends, relevant to the subject matter. | 3.2386 | Agree |
| 6. Demonstrate first the activity | 3.2500 | Strongly Agree |
| 7. Is organized in exercises designed to reinforce content. | 3.1477 | Agree |
| 8. Gives and explains rubric in evaluating the activity. | 3.2273 | Agree |
| 9. Uses various strategies, teaching aids/devices, techniques in presenting the lessons. | 3.1705 | Agree |
| 10. Is organized in presenting subject matters by systematically following course outline. | 3.1818 | Agree |
| 11. Has mastery of the subject matter. | 3.1705 | Agree |
| **OVERALL** | 3.2035 | Agree |

As shown in Table 2.4, the statement with the highest mean of 3.2500 is “Demonstrate first the activity.” It shows that the level of Academic Engagement of the Cookery students in terms of teacher-related aspect is highly encouraged and often present. This reflects a result that teachers’ demonstration of activities strongly enhances students’ understanding and engagement in the TVL track.

The finding from Table 2.4 highlights that students highly value it when teachers demonstrate an activity first, which makes perfect sense because

watching a cooking process step-by-step is much easier to grasp than just reading a recipe. To back this up, a recent study by Pasquito and Singzon (2025) looked exactly at how guided demonstrations affect senior high school students in the TVL Cookery track, proving that when teachers physically show the techniques before letting students try, it successfully bridges the gap between textbook ideas and actual kitchen work. Because the teacher models the right way to do the task first, students know exactly what the final product should look like and feel much less stressed about making mistakes. The researchers noted that this hands-on teaching style directly leads to higher student engagement, more confidence, and much better practical test scores, which strongly supports your conclusion that teacher demonstrations are a key part of keeping cookery students focused and helping them truly understand their lessons.

It is also evident in Table 2.4 that the statement, “Is organized in exercises designed to reinforce content,” has the lowest mean of 3.1477, which is described as agree and is sometimes present. The result reflects a conclusion that organized exercises still encourage students’ Academic Engagement in terms of teacher-related aspect, although its influence is slightly less compared to other statements under the same indicator.

The finding from Table 2.4 shows that while organizing exercises to reinforce lessons is helpful, it has a slightly lower impact on student engagement compared to other teacher actions. This is completely normal, as repetitive practice can sometimes feel like a routine chore rather than an exciting activity. To back this up, a recent study by Yuldasheva (2025) found that while traditional reinforcement

exercises are necessary for building foundational skills, students often view them as the least engaging part of their coursework. The research points out that because structured drills can easily become monotonous, they simply do not capture a learner's attention as strongly as active, hands-on tasks or live demonstrations do. However, because these exercises are still crucial for locking in the lesson, they continue to positively support the students' overall learning and focus, which perfectly explains why your respondents still "agreed"

with the statement despite it having the lowest mean.

*The Level of Academic Engagement of Cookery Students in terms of On-Instructional Materials*

The following Table 2.5 indicates The Level of Academic Engagement of Cookery Students in terms of On-Instructional Materials. There are 7 statements in it, each having a meaning and description that corresponds to it.

*Table 2.5. The Level of Academic Engagement of Cookery Students in terms of On-Instructional Materials*

| On-Instructional Materials | Mean | Description |
|---|---|---|
| 1. Uses PowerPoint presentation to explain the lesson. | 3.4091 | Strongly Agree |
| 2. Uses resourceful tools and equipment for project development. | 3.2841 | Strongly Agree |
| 3. Present the tools and equipment needed for the activity. | 3.3295 | Strongly Agree |
| 4. Arrangements to obtain materials were made. | 3.2273 | Agree |

| | | |
|---|---|---|
| 5. Introduce special equipment/supplies needed. | 3.2500 | Strongly Agree |
| 6. Video presentation showing about the lesson. | 3.2273 | Agree |
| 7. Chalks and blackboards in explaining the lesson. | 3.1477 | Agree |
| **OVERALL** | 3.2679 | Agree |

As shown in Table 2.5, the statement with the highest mean of 3.4091 is "Uses PowerPoint presentation to explain the lesson." It shows that the level of Academic Engagement of the Cookery students in terms of on-instructional materials is highly encouraged and often present. This reflects a result that the use of PowerPoint presentations effectively enhances students' understanding and engagement in the TVL track.

The finding from Table 2.5 shows that students are highly engaged when teachers use PowerPoint presentations to explain their lessons. This makes complete sense because visual aids help break down complex information into easy-to-understand pieces, especially in a hands-on course like Cookery. To back this up, a recent study by Mardianto et al. (2022) found that active student engagement via PowerPoint directly enhances academic performance and creates a highly favorable attitude toward learning. When teachers use slide presentations, they can include photos, step-by-step bullet points, and clean visuals that make the lesson much more interesting than just listening to a standard lecture. Because

these visual elements easily capture the learners' attention, students are able to grasp the procedures and theories much faster. This perfectly supports your result that using PowerPoint is a highly effective way to keep TVL Cookery students motivated and fully engaged with their instructional materials.

As shown in Table 2.5, the statement with the lowest mean of 3.1477 is “Chalks and blackboards in explaining the lesson.” It shows that the level of Academic Engagement of the Cookery students in terms of on-instructional materials is encouraged but to a lesser extent compared to other materials. This reflects a result that while students still engage with lessons delivered through traditional chalk and blackboard methods, these tools are less effective in fully capturing students’ attention and promoting active participation in the TVL track.

The finding from Table 2.5 shows that while chalk and blackboards are still used, they are the least effective at keeping students fully engaged compared to modern tools. This makes perfect sense because traditional boards only offer plain text and simple drawings, which can be a bit boring for a highly visual and hands-on subject like Cookery. To back this up, a recent study by Kharismaputra et al. (2020) found that traditional teaching tools often struggle to sustain learner interest in vocational education because they lack dynamic, real-world visual representation. Their research emphasized that while the chalkboard is reliable for basic explanations, it simply cannot provide the realistic, step-by-step imagery needed for practical courses. Because students in the TVL track need to see actual colors, textures, and clear procedures to fully grasp culinary techniques, the plain chalkboard cannot provide the same level of excitement or clarity as digital

presentations or videos. This strongly supports your conclusion that while traditional boards still work to some extent, they are just not the best option for maximizing student participation and focus in a modern Cookery class.

*Significance on the Relationship between Student Preferences on TVL Track and Academic Engagement*

Table 4 presents the significance of correlation tests. The table displays the r-value, p-value, and the decision on hypothesis is shown in the table.

*Table 4. Significance on the Relationship between Artificial Intelligence and Academic Performance of the Senior High School Students*

| | **Relationship between Students Preferences and Academic Engagement** | | | |
|---|---|---|---|---|
| | r value | p value | Decision on $H_0$ | Interpretation |
| **Personal-Related Factors** | 0.8084 | p<0.001 | Reject | **Significant relationship** |

r-value = 0.8084, very strong positive correlation

p-value<0.001, reject the null hypothesis

It can be drawn in Table 4 that there is a significant relationship between personal-related factors and academic engagement of the Cookery students, with p-value (p<0.001) which is less than 0.05. With this result, the null hypothesis is therefore rejected. The researchers can also see that the variables in the first and second statements of the problem statistically correlate with one another.

Additionally, the researchers can conclude that personal-related factors and academic engagement are related. There is a very strong positive correlation with an r value of 0.8084. This describes how students' personal-related factors greatly influence their level of academic engagement in the TVL track.

In essence, this conclusion is supported by existing studies indicating that students' preferences and motivational factors have a significant relationship with their academic engagement. In a study conducted in the United States, it was found that students who exhibit higher levels of personal interest and self-related motivations demonstrate greater behavioral and emotional engagement in academic tasks, which in turn enhances their overall learning experience (Wang et al., 2020). These findings align with the present study, where students' preferences toward the TVL track, particularly in terms of personal-related factors, contribute to their level of active participation and involvement in learning activities. Furthermore, the results revealed that students' preferences are encouraged, while their academic engagement is also evident, indicating that stronger preferences lead to higher engagement. This supports the conclusion that there is a very strong positive relationship between students' preferences and academic engagement.

# Chapter 4

## Conclusions and Recommendations

The conclusion of this study consists of a concise summary of the findings. After analyzing the interpreted data, this section gives the conclusions, which summarize the determined primary results, and recommendations for the beneficiaries of this study.

### Conclusions

Based on the data, the study found a very strong positive and significant correlation between Student Preferences on TVL Track and Academic Engagement. Hence, the study rejected the null hypothesis. The findings suggest that a student's personal preference for the TVL track serves as a massive driving factor that positively affects their academic engagement. Because the students genuinely prefer and have a personal interest in the Cookery track, they are naturally far more motivated to participate, sustain their focus, and perform well in their hands-on and theoretical activities. The results also indicate that the indicators for Academic Engagement such as “Study Habits,” “Academic Ability,” “Teacher-Related Aspect,” and “On-Instructional Materials” all have high descriptive values. The interpretation of these indicators is highly encouraged and often present. Additionally, the overall variable of Academic Engagement has a very satisfactory descriptive value. Overall, it means that this high level of active participation and focus is consistently evident among the Cookery students.

## Recommendations

The researchers make the following recommendations based on the findings and inferences made from the study.

First, the Department of Education (DepEd) should continue developing policies that guide junior high school students in discovering their true personal interests before selecting the TVL Cookery strand, given the study's finding that a genuine passion for the culinary arts has a very strong positive relationship with academic engagement. Policies should also focus on allocating specific funds to upgrade Cookery instructional technologies, ensuring culinary laboratories have access to modern multimedia tools—such as digital screens for recipe demonstrations—rather than relying solely on traditional chalkboards.

Second, school administrators should create an environment that directly supports hands-on culinary learning by upgrading and strictly maintaining specialized Cookery laboratory facilities. Ensuring these laboratory spaces mirror professional, orderly commercial kitchens will support the Cookery students' proven need for a clean, highly structured environment. Administrators should also encourage the use of modern visual aids by equipping kitchen-classrooms with safe, wall-mounted projectors and screens so teachers can effectively present visual lectures alongside their practical cooking sessions.

Third, teachers play a crucial role in sustaining the high engagement levels of TVL Cookery students. Since the findings highlight the

effectiveness of visual and practical instruction, Cookery instructors should prioritize step-by-step "guided demonstrations" of recipes and cooking techniques before asking students to perform practical kitchen assessments. Furthermore, teachers should transition away from traditional chalk and blackboards, utilizing highly visual PowerPoint presentations and video modeling to break down complex food preparation procedures, safety standards, and plating techniques to keep students actively focused.

Fourth, parents should actively support their children’s education by helping them establish a specific, clean, and orderly study and practice space at home. Since an organized environment was proven to significantly boost a Cookery student's focus and academic engagement, parents can greatly contribute to their child's practical success by minimizing household distractions, providing a dedicated space for them to practice basic skills like recipe planning, and continually validating their personal choice to pursue the culinary arts.

Fifth, TVL Cookery students should take initiative in their learning by consciously maintaining strict sanitation and highly organized work habits both at their home study spaces and at their laboratory workstations (practicing proper *mise en place*). Because personal preference is a major driver of their engagement, students should continually align their daily kitchen activities and practical exercises with their long-term culinary passions, while ensuring they balance extracurricular activities so their focus on theoretical and practical Cookery assessments is not hampered.

Lastly, future researchers should build upon these findings by exploring how the strong relationship between personal preference and academic engagement evolves among TVL Cookery students as they progress to Grade 12 and prepare for their National Certificate (NC II) assessments. Expanding the sample to include Cookery students across different schools or districts could provide a more comprehensive view of how specific instructional materials, commercial kitchen environments, and live teacher demonstrations impact long-term culinary performance,

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